Forecasting Government Expenditure in Ghana

Using Autoregressive Integrated Moving Average
with Exogenous Variables (ARIMAX)

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Abstract: This study develops and evaluates an Autoregressive Integrated Moving Average with Exogenous Variables (ARIMAX) model for forecasting Ghana's government expenditure over the period 1984–2024. The dependent variable is log-transformed total government expenditure (USD equivalent), and three exogenous variables are examined — population, log government revenue, and log GDP — both jointly and in isolation. The ARIMAX framework is compared against a univariate ARIMA baseline using information criteria (AIC, BIC) and standard accuracy metrics (RMSE, MAE, MAPE, MASE). Results demonstrate that incorporating macroeconomic exogenous variables into the time series framework meaningfully improves forecasting performance. Policy implications for Ghana's medium-term fiscal planning are discussed.

1 Chapter One: Introduction

1.1 Background of Study

Government expenditure is a critical pillar of Ghana's development, financing key sectors such as education, healthcare, infrastructure, and social protection that contribute directly to productivity growth, human capital development, and long-term economic performance (World Bank, 2023). In the majority of developing countries, government is seen as an instrument of change, and hence the size of government expenditure reveals the magnitude of government involvement in the economy.

The shift in the role of government — from traditional functions such as provision of security, administration, and law and order, to direct intervention in income-generating activities like capital investment and distributive roles like subsidies and transfers — has significantly expanded the scope of governments in many countries across the globe.

Government expenditure is defined as the total spending by the government on recurrent items and capital projects, primarily funded through budgetary appropriations and often involving significant allocations for personnel costs (Leonard Onyiruba, 2016). In Ghana, government expenditure includes various components such as total public spending on purchase of goods and services, compensation of employees, interest payments, grants to other government units, non-financial assets, and capital expenditure. As defined by the

International Monetary Fund (IMF), general government expenditure consists of total expense and the net acquisition of nonfinancial assets (Statista, 2025).

Ghana's fiscal exposure is characterised by its status as a commodity-dependent economy. Changes in world prices of cocoa, gold, and oil directly affect export earnings, tax revenues, and the overall fiscal balance. When commodity prices rise, revenue and spending pressures typically rise; when prices fall, fiscal stress often triggers spending cuts or arrears. Ghana's public spending has fluctuated significantly due to changes in political priorities, inflationary pressures, and debt servicing obligations — inconsistencies that make it difficult for policymakers to predict expenditure accurately, often resulting in fiscal imbalances and budget deficits.

These trends underscore a critical need for reliable forecasting tools that can help policymakers anticipate future expenditure paths, identify potential fiscal risk, and make informed, data-driven decisions that ensure long-term fiscal sustainability.

Traditional estimation methods have limitations in handling dynamic economic data; hence time-series models such as the **ARIMAX** (Autoregressive Integrated Moving Average with Exogenous Variables) offer a more reliable approach. In many real situations, the assumption that only past values influence future outcomes is too limited because external factors often play a significant role in influencing the direction and behaviour of a time series. This is where ARIMAX becomes particularly valuable — by integrating exogenous variables directly into the forecasting process, allowing for more realistic and accurate predictions.

1.2 Problem Statement

Government expenditure in Ghana faces several significant challenges and limitations, primarily driven by high public debt, a narrow revenue base, and weaknesses in public financial management (PFM) systems. Despite fiscal reforms, Ghana continues to face budget overruns, expenditure volatility, and debt-related pressures. The May 2023 IMF Extended Credit Facility (ECF) arrangement was triggered by severe fiscal and debt vulnerabilities following external shocks and policy slippages (Ministry of Finance, Ghana).

Awariefe & Ogumeyo (2023) found that traditional regression (MLR) underperforms when predicting government expenditure because it cannot handle complex, non-linear macro dynamics. Djakou & Jiang (2023) found that an ARIMA(1,1,1) model assumes a linear trend and does not fully capture external macro shocks such as volatility in budget performance. While several studies have applied ARIMA-type models in Ghana for forecasting inflation, GDP, or tax revenue, there is limited evidence of similar applications

for **government expenditure forecasting using the ARIMAX framework**.

This methodological gap reduces policymakers' ability to anticipate the implications of macroeconomic shocks and IMF fiscal regimes. This study addresses the gap by applying ARIMAX with population, revenue, and GDP as exogenous variables to deliver more reliable fiscal forecasts for Ghana.

1.3 Objectives of the Study

General Objective: To develop an ARIMAX-based forecasting model for Ghana's government expenditure, using population, government revenue, and GDP as exogenous variables.

Specific Objectives:

1. To examine historical trends and volatility in Ghana's government expenditure (1983–2024).
2. To assess the influence of macroeconomic variables (population, revenue, GDP) on government expenditure.
3. To estimate optimal ARIMAX specifications and generate out-of-sample forecasts.
4. To evaluate whether ARIMAX with exogenous variables improves forecasting accuracy relative to a univariate ARIMA baseline.

1.4 Research Questions

1. What are the dominant trends and fluctuations in Ghana's government expenditure over time?
2. How do population growth, government revenue, and GDP movements affect government expenditure?
3. Does an ARIMAX model with macroeconomic exogenous variables improve forecasting accuracy relative to ARIMA?
4. Which combination of exogenous variables yields the most parsimonious and accurate ARIMAX specification?

1.5 Significance of the Study

This study is significant on multiple fronts:

- **Policy relevance:** Improves expenditure forecasts that help the Ministry of Fi-

nance and Parliament set realistic budget ceilings, reduce expenditure arrears, and better manage fiscal risks.

- **Methodological contribution:** Demonstrates the effectiveness of parsimonious ARIMAX modelling in a developing, commodity-dependent economy, contributing to the applied time-series econometrics literature.
- **Institutional utility:** Provides a replicable, transparent, and data-driven analytical framework that can be adopted by the Bank of Ghana, the National Development Planning Commission, and development partners for medium-term expenditure planning.

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2 Chapter Two: Literature Review

2.1 Theoretical Foundations

The theoretical framework for this study draws on several interrelated strands of economic and statistical theory:

Box-Jenkins ARIMA Methodology
The theoretical foundation for ARIMA (Autoregressive Integrated Moving Average) modelling is based on the Box-Jenkins methodology developed by George Box and Gwilym Jenkins (1970), which establishes a systematic process for identification, estimation, and diagnostic checking of time series models. ARIMAX extends this framework by incorporating exogenous regressors (transfer functions), allowing the model to capture external influences on the primary series.

Keynesian Fiscal Theory
John Maynard Keynes (1936) posits that government spending is a critical component of aggregate demand and a powerful tool for economic stabilisation. This perspective reinforces the importance of accurately forecasting expenditure to guide effective fiscal planning and ensure macroeconomic stability. According to Keynesian theory, an increase in government expenditure leads to an increase in aggregate demand, production, and ultimately GDP — making GDP a natural exogenous candidate in the ARIMAX framework.

Wagner's Law
Named after German economist Adolph Wagner, this hypothesis posits that as an economy develops over time, government expenditure grows more than proportionally relative to

national income. Wagner's Law implies a long-run upward drift in public spending as economies grow, providing the theoretical basis for including GDP and population as exogenous drivers of expenditure.

Peacock-Wiseman Displacement Hypothesis
Peacock and Wiseman (1961) argue that public expenditure does not increase at a steady rate, but rather jumps in response to major economic or social disturbances (wars, crises, pandemics). Once elevated, spending rarely returns to its pre-shock level — a "displacement" effect. This theoretical lens is important for understanding Ghana's expenditure jumps around periods of economic crisis.

Commodity-Price Fiscal Linkage
In commodity-exporting economies, global price shocks influence tax receipts, borrowing costs, and expenditure choices. IMF working paper evidence (Spatafora & Samake, 2012) shows both revenue and expenditure respond significantly to commodity price movements, making fiscal policy highly procyclical in such settings.

2.2 Empirical Literature

A growing body of literature confirms that ARIMAX models consistently outperform baseline ARIMA models when relevant exogenous information is available:

- **Jiayi Yang (2025):** Applied ARIMA, ARIMAX, VAR, and GLM to forecast China's university admission rates (1977–2024). The ARIMAX model, incorporating GDP and newborn population as exogenous variables, produced the lowest forecast errors and provided deeper insights into how macroeconomic factors drive admission rates.
- **Fairuz Azizah et al. (2025):** Modelled Indonesia's Human Development Index using ARIMAX with inflation and infant mortality as exogenous variables. Results confirmed that including these external drivers significantly improved estimation accuracy over the pure ARIMA specification.
- **Coker (2025):** Compared ARIMA, ARIMAX, and VAR for short-term inflation forecasting in Sierra Leone. The ARIMAX model incorporating the exchange rate as an exogenous variable significantly outperformed the alternatives, underscoring the importance of external drivers in small, open economies.
- **Abdulazeez (2021):** Demonstrated in a Nigeria GDP forecasting study that ARIMAX with agriculture, industry, and services sector shares consistently outperformed ARIMA across all error metrics (RMSE: 899 vs 1,662; MSE: 568 vs 1,161).

- **Ajibode & Adeboye:** Applied ARIMA and ARIMAX to COVID-19 mortality modelling in Nigeria. ARIMAX with confirmed cases as an exogenous variable produced the lowest MAE (5.98) and RMSE (8.54), outperforming the pure ARIMA specification.
- **Michael Safo Ofori et al.:** Forecasting VAT revenue in Ghana using ARIMA with intervention, demonstrating the utility of time-series methods for Ghanaian fiscal data and validating the approach adopted in the present study.
- **Boachie et al. (2022):** Examined Ghana's public health spending (1980–2014) using 3SLS estimators, finding that IMF programme conditionalities slightly constrained public health spending, supporting the inclusion of IMF regime indicators in fiscal forecasting models.

2.3 Chapter Summary

The reviewed literature consistently highlights that: (1) ARIMA models are a well-established baseline for univariate fiscal forecasting; (2) the incorporation of exogenous variables through the ARIMAX framework reliably improves forecasting accuracy; (3) macro-fiscal variables such as GDP, revenue, and population are the most empirically validated exogenous drivers; and (4) there is a notable gap in the application of ARIMAX to Ghana's government expenditure forecasting — the gap this study addresses.

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3 Chapter Three: Methodology

3.1 Research Design

A **quantitative, predictive time-series design** is adopted, consistent with the Box-Jenkins tradition. The study employs secondary annual data covering the period **1983–2024** (42 observations), drawing on official government fiscal records and national accounts. The research is both explanatory (examining the influence of macroeconomic exogenous variables on expenditure) and predictive (generating a 10-year out-of-sample forecast).

3.2 Data Sources and Variables

Table 1: Table 1: Variable Description and Sources

Variable	Description	Unit	Source
GE (Y)	Total Government Expenditure (Net Lending)	Million GHS → log(USD)	Bank of Ghana — Gov. Fiscal Operations
POPULATION (X1)	Ghana's Annual Population	Persons	World Bank / Ghana Statistical Service
REVENUE (X2)	Total Government Revenue (Million GHS → USD)	log(USD)	Bank of Ghana — Gov. Fiscal Operations
GDP (X3)	Nominal GDP (USD Billion → USD)	log(USD)	World Bank National Accounts

The dataset spans **1983 to 2024**, with the following coverage:

- **Government Expenditure:** Annual total expenditure and net lending (millions of Ghana Cedis), sourced from the Bank of Ghana's Government Fiscal Operations report.
- **Population:** Annual population figures used as a demographic driver of public expenditure demand.
- **Revenue:** Annual government revenue (millions of Ghana Cedis), converted to USD and log-transformed.
- **GDP:** Nominal GDP in USD billions, converted to USD and log-transformed.

All currency-denominated variables (expenditure, revenue) are **converted from Ghana Cedis to US Dollars** using the contemporaneous annual average exchange rate. This conversion normalises the data against the extreme nominal exchange-rate depreciation that Ghana experienced over the study period, particularly post-2014. **Log transformations** are then applied to all monetary variables to stabilise variance and linearise exponential growth trends.

3.3 Analytical Framework: The ARIMAX Model

3.3.1 ARIMA Components

The ARIMA(p, d, q) model combines three core components:

- **AR(p) — Autoregressive component:** Models the relationship between the

current observation and (p) lagged values:

$$X_t = c + \phi_1 X_{t-1} + \phi_2 X_{t-2} + \ldots + \phi_p X_{t-p} + \epsilon_t$$

- **I(d) — Integration (Differencing):** The series is differenced (d) times to achieve stationarity, removing trend and unit-root behaviour.
- **MA(q) — Moving Average component:** Models the relationship between the current value and (q) lagged forecast errors:

$$X_t = c + \epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \ldots + \theta_q \epsilon_{t-q}$$

Combined:

$$X_t = c + \sum_{i=1} p\phi_i X_{t-i} + \sum_{j=1} q\theta_j \epsilon_{t-j} + \epsilon_t$$

3.3.2 ARIMAX Extension

The ARIMAX model augments the ARIMA framework with exogenous regressors X_k:

$$Y_t = \mu + \sum_{i=1} p\phi_i Y_{t-i} + \sum_{j=1} q\theta_j \epsilon_{t-j} + \sum_k \beta_k X_{k,t} + \epsilon_t$$

where: - Y_t = Government Expenditure (log USD) at time t - $X_{k,t}$ = exogenous variables (population, log revenue, log GDP) - β_k = coefficients for the exogenous variables - $\epsilon_t \sim WN(0, \sigma 2)$

3.3.3 Model Selection Criteria

Criterion	Formula	Preference
AIC	$-2\log(L) + 2k$	Lower is better
BIC	$-2\log(L) + k\log(n)$	Lower is better
RMSE	$\sqrt{\frac{1}{n}\sum e_t 2}$	Lower is better
MAE	$\frac{1}{n}\sum \lvert e_t \rvert$	Lower is better
MAPE	$\frac{100}{n}\sum \lvert e_t Y_t \rvert$	Lower is better
MASE	Scaled by naïve forecast MAE	< 1 is better

3.3.4 Stationarity Tests

Augmented Dickey-Fuller (ADF) Test:

$$\Delta y_t = \alpha + \beta t + \rho y_{t-1} + \sum_{i=1} k\phi_i \Delta y_{t-i} + \epsilon_t$$

- H_0: Unit root present (non-stationary)
- H_1: Series is stationary
- Reject H_0 if p-value < 0.05

---

4 Chapter Four: Data Analysis and Results

4.1 Setup — Libraries and Global Options

All required R packages are loaded here. These cover time series modelling (`forecast`, `tseries`), multicollinearity diagnostics (`car`), data import (`readxl`), and data manipulation/visualisation (`tidyverse`).

```
# Time series modelling and forecasting toolkit
library(forecast)

# Augmented Dickey-Fuller and KPSS unit root tests
library(tseries)

# Variance Inflation Factor for multicollinearity diagnostics
library(car)

# Reading Excel data files
library(readxl)

# Data manipulation and visualisation (dplyr, ggplot2, tidyr)
library(tidyverse)
```

4.2 Data Loading and Preprocessing

4.2.1 Loading the Dataset

```
# Load the annual fiscal dataset
data_raw <- read_excel("Data2.xlsx")

# Preview the raw data structure
glimpse(data_raw)
```

```
Rows: 42
Columns: 7
$ YEAR                        <dbl> 1983, 1984, 1985, 1986, 1987, 1988, 1989, 1990, 1991, 1992, ....
$ EXPENDITURE                 <dbl> 1.51, 2.75, 4.79, 7.33, 10.69, 14.99, 20.42, 26.40, 35.16, ...
$ POPULATION                  <dbl> 13029836, 13370590, 13661229, 13965217, 14289108, 14636568, ...
$ ELECTION                    <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0, 0, 1, 0, 0,...
$ `GDP USD BILLION`           <dbl> 4.06, 4.41, 4.50, 5.74, 5.07, 5.20, 5.25, 5.89, 6.60, ...
$ `REVENUE MILLION OF  CEDIS` <dbl> 1.03, 2.26, 4.03, 7.36, 11.10, 15.38, 21.45, 26.73, ...
$ EXCHANGE_RATE               <dbl> 0.00, 0.01, 0.01, 0.01, 0.02, 0.02, 0.03, 0.03, 0.04, 0.04, ...
```

4.2.2 Currency Conversion and Log Transformation

Government expenditure and revenue are denominated in **millions of Ghana Cedis (GHS)**. To control for the severe nominal depreciation of the cedi over the study period — from roughly 2.7 GHS/USD in 1984 to over 12 GHS/USD by the early 2020s — all monetary variables are **converted to US Dollars** using the contemporaneous annual exchange rate.

Log transformations are applied to all monetary variables to: 1. Stabilise the exponential growth in variance (heteroscedasticity correction) 2. Linearise the long-run upward trend, making the series more amenable to ARIMA modelling 3. Enable percentage-change interpretation of first differences

```
data <- data_raw |>
mutate(
# Convert GHS expenditure (millions) to USD
GOV_EXP_DOLLARS = EXPENDITURE * 1000000 / EXCHANGE_RATE,

# Convert GHS revenue (millions) to USD
REVEVENUE = `REVENUE MILLION OF  CEDIS` * 1000000 /
   EXCHANGE_RATE,
```

```r
# Convert GDP (USD billions) to USD for unit consistency
GDP = `GDP USD BILLION` * 1000000000,

# Apply natural log transformation to all monetary variables
# Rationale: log stabilises exponential variance growth and
   linearises trend
LOG_EXPENDITURE = log(GOV_EXP_DOLLARS),
LOG_REVENUE     = log(REVEVENUE),
LOG_GDP         = log(GDP)
) |>
# Remove original pre-transformation columns and the election
   dummy
# (election effects are not part of this specification)
select(-c(EXPENDITURE, `REVENUE MILLION OF  CEDIS`, `GDP USD
   BILLION`,
EXCHANGE_RATE, ELECTION)) |>
# Remove rows where division by zero produced Inf values
filter(GOV_EXP_DOLLARS != Inf)

# Show the processed dataset structure
glimpse(data)
```

```
Rows: 38
Columns: 8
$ YEAR            <dbl> 1984, 1985, 1986, 1987, 1988, 1989, 1990, 1991, 1992,...
$ POPULATION      <dbl> 13370590, 13661229, 13965217, 14289108, 14636568,...
$ GOV_EXP_DOLLARS <dbl> 275000000, 479000000, 733000000, 534500000, ...
$ REVEVENUE       <dbl> 226000000, 403000000, 736000000, 555000000, ...
$ GDP             <dbl> 4.410e+09, 4.500e+09, 5.740e+09, 5.070e+09,...
$ LOG_EXPENDITURE <dbl> 19.43228, 19.98721, 20.41266, 20.09684, ...
$ LOG_REVENUE     <dbl> 19.23605, 19.81445, 20.41674, 20.13448,...
$ LOG_GDP         <dbl> 22.20714, 22.22734, 22.47073, 22.34661, ...
```

4.2.3 Processed Data Summary

```r
summary(data[, c("GOV_EXP_DOLLARS", "LOG_EXPENDITURE",
"LOG_REVENUE", "LOG_GDP", "POPULATION")])
```

```
GOV_EXP_DOLLARS     LOG_EXPENDITURE  LOG_REVENUE       LOG_GDP        POPULATION
Min.   :2.750e+08   Min.   :19.43   Min.   :19.24   Min.   :22.21   Min.   :13370590
1st Qu.:1.299e+09   1st Qu.:20.98   1st Qu.:20.71   1st Qu.:22.50   1st Qu.:16690437
```

```
Median :2.044e+09   Median :21.44   Median :21.24   Median :22.76   Median :21013046
Mean   :5.374e+09   Mean   :21.79   Mean   :21.60   Mean   :23.38   Mean   :21768557
3rd Qu.:9.750e+09   3rd Qu.:23.00   3rd Qu.:22.86   3rd Qu.:24.43   3rd Qu.:26574473
Max.   :1.881e+10   Max.   :23.66   Max.   :23.21   Max.   :25.10   Max.   :32518665
```

4.2.4 Defining Time Series Objects

```r
# Dependent variable: log government expenditure as annual
   time series
# Starting from 1984 (first year with valid exchange rate data)
Y <- ts(data$LOG_EXPENDITURE, start = 1984, frequency = 1)

# Exogenous variable matrix used in ARIMAX models
xreg_vars <- data[, c("POPULATION", "LOG_REVENUE", "LOG_GDP")]

# Forecast horizon: 10 years beyond the end of the estimation
   sample
h_period <- 10

cat("Dependent variable (Y): Log Government Expenditure\n")
cat("Sample period:", start(Y)[1], "to", end(Y)[1], "\n")
cat("Observations:", length(Y), "\n")
cat("Forecast horizon:", h_period, "years\n")
```

```
Dependent variable (Y): Log Government Expenditure
Sample period: 1984 to 2021
Observations: 38
Forecast horizon: 10 years
```

4.3 Descriptive Analysis and Trend Exploration

4.3.1 Historical Trend of Government Expenditure

Before formal modelling, it is important to visually inspect Ghana's government expenditure history. The time series plot below reveals the character of the data: the direction of the trend, any structural breaks, and the general growth trajectory.

```r
par(mfrow = c(2, 1), mar = c(4, 5, 3, 2))
```

```r
# Plot 1: Raw USD expenditure
plot(ts(data$GOV_EXP_DOLLARS / 1e9, start = 1984),
type = "o", col = "steelblue", lwd = 2,
main = "Ghana Total Government Expenditure (USD Billion)",
ylab = "USD Billion", xlab = "Year")
grid(col = "lightgray", lty = "dotted")

# Plot 2: Log-transformed expenditure (used as Y in models)
plot(Y,
type = "o", col = "darkgreen", lwd = 2,
main = "Log Government Expenditure (used as dependent variable
   Y)",
ylab = "Log(USD)", xlab = "Year")
grid(col = "lightgray", lty = "dotted")
```

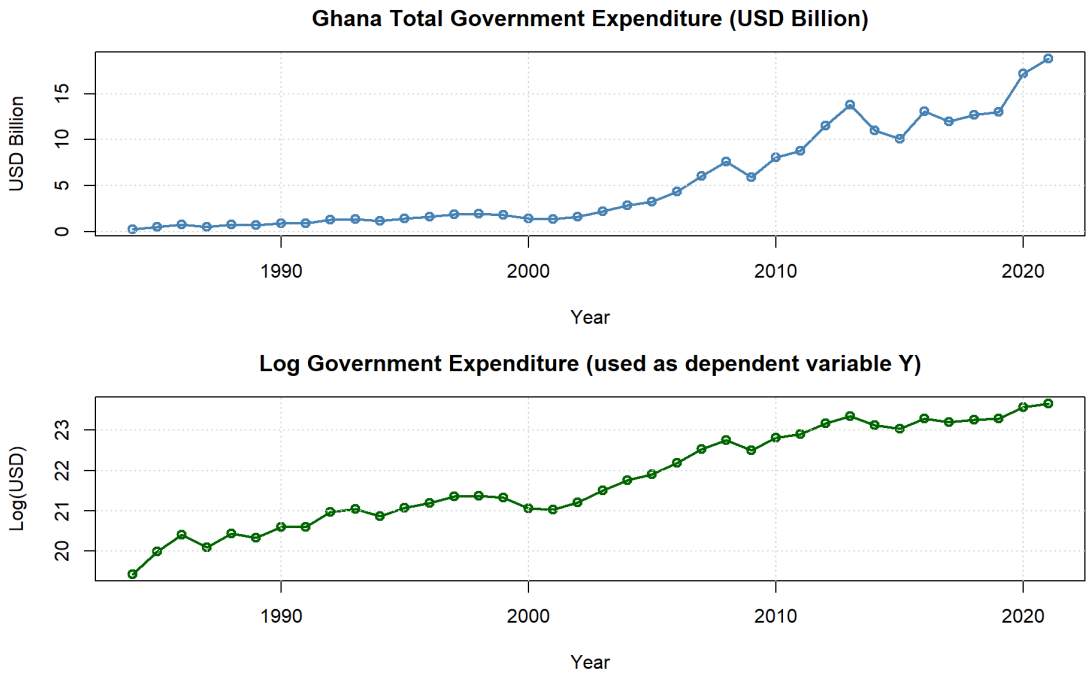

Figure 1: Ghana's Log Government Expenditure in USD (1984–2024)

Figure 1: Ghana's Log Government Expenditure in USD (1984–2024)

Observation: The raw expenditure series exhibits exponential growth — characteristic of a nominal fiscal variable in a developing economy undergoing both real growth and significant price-level increases. The log transformation

converts this to a near-linear upward trend, more appropriate for ARIMA-based modelling. Notable inflections (post-2014 cedi depreciation, post-2020 COVID-19 fiscal expansion) are visible in both plots.

4.3.2 Exogenous Variables Over Time

```
par(mfrow = c(3, 1), mar = c(3, 5, 2, 2))

plot(ts(data$POPULATION / 1e6, start = 1984),
type = "o", col = "darkorange", lwd = 1.5,
main = "Ghana Population (Millions)", ylab = "Millions", xlab
   = "")
grid(col = "lightgray", lty = "dotted")

plot(ts(data$LOG_REVENUE, start = 1984),
type = "o", col = "purple", lwd = 1.5,
main = "Log Government Revenue (USD)", ylab = "Log(USD)", xlab
   = "")
grid(col = "lightgray", lty = "dotted")

plot(ts(data$LOG_GDP, start = 1984),
type = "o", col = "firebrick", lwd = 1.5,
main = "Log GDP (USD)", ylab = "Log(USD)", xlab = "Year")
grid(col = "lightgray", lty = "dotted")
```

Observation: All three exogenous variables exhibit persistent upward trends over the sample period. Population growth is smooth and demographic in nature. Revenue and GDP share a similar trajectory to expenditure, reflecting the co-movement between fiscal variables and macroeconomic performance over Ghana's development path. This co-movement is precisely the signal that ARIMAX is designed to exploit.

4.4 Multicollinearity Check (VIF)

Before including multiple exogenous regressors in the ARIMAX model, potential **multicollinearity** is assessed using the **Variance Inflation Factor (VIF)**. Multicollinearity

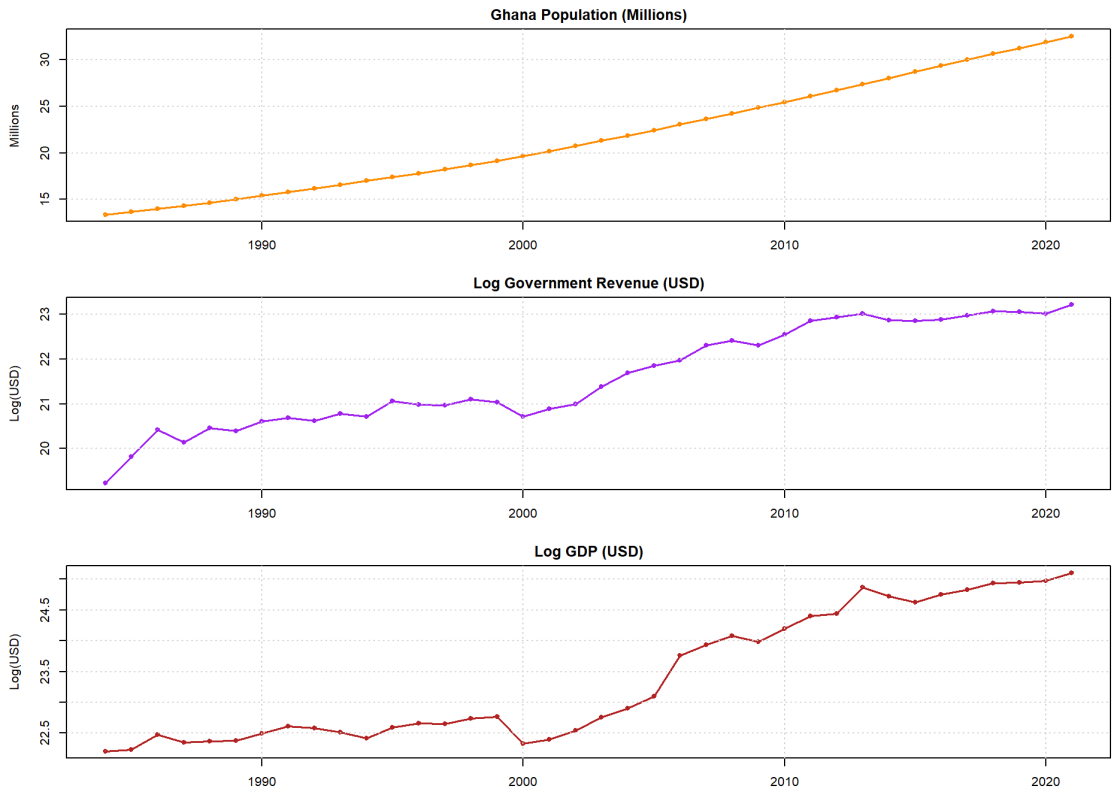

Figure 2: Exogenous Variables: Population, Log Revenue, and Log GDP (1984–2024)

arises when two or more predictors are highly linearly correlated, inflating coefficient standard errors and undermining interpretability.

Decision rule: - VIF $< 5 \rightarrow$ Acceptable - VIF 5–10 $\rightarrow$ Moderate concern; monitor - VIF $> 10 \rightarrow$ Severe multicollinearity $\rightarrow$ consider dimensionality reduction or variable exclusion

Given that population, revenue, and GDP are all trending upward over the same period, some degree of multicollinearity is expected. This motivates the individual ARIMAX models (one exogenous variable at a time) fitted in Section 4.7–4.9.

```
# Fit auxiliary OLS regression solely to compute VIF values
# (Not used for forecasting; diagnostic purpose only)
vif_model <- lm(Y ~ POPULATION + LOG_REVENUE + LOG_GDP, data =
   data)

cat("=== Variance Inflation Factors (VIF) ===\n")
print(vif(vif_model))

cat("\nNote: VIF > 10 indicates severe multicollinearity.\n")
cat("VIF > 5 suggests moderate collinearity that may inflate
```

```
    standard errors.\n")
```

```
=== Variance Inflation Factors (VIF) ===
POPULATION LOG_REVENUE     LOG_GDP
17.69051    20.10369    15.43043

Note: VIF > 10 indicates severe multicollinearity.
VIF > 5 suggests moderate collinearity that may inflate standard errors.
```

```
cat("=== ADF Test: Log Government Expenditure (Level) ===\n")
adf_level <- adf.test(Y)
print(adf_level)

if (adf_level$p.value > 0.05) {
  cat("\n>> Decision: Fail to reject H0 (p =",
     round(adf_level$p.value, 4),
  ")     Series is NON-STATIONARY. Differencing required.\n")
} else {
  cat("\n>> Decision: Reject H0 (p =",
     round(adf_level$p.value, 4),
  ")     Series is STATIONARY. No differencing needed.\n")
}
```

```
=== ADF Test: Log Government Expenditure (Level) ===

Augmented Dickey-Fuller Test

data:  Y
Dickey-Fuller = -2.2832, Lag order = 3, p-value = 0.4626
alternative hypothesis: stationary


>> Decision: Fail to reject H0 (p = 0.4626 ) - Series is NON-STATIONARY. Differencing required.
```

Interpretation: High VIF values among `LOG_REVENUE`, `LOG_GDP`, and `POPULATION` are plausible given that all three grow in tandem with the Ghanaian economy over time. The joint ARIMAX model (Section 4.6) is included for completeness, but the individual models (Sections 4.7–4.9) provide cleaner, more interpretable estimates by avoiding multicollinear combinations.

4.5 4.5 Stationarity Testing and Order of Integration

4.5.1 ADF Test on the Level Series

The ADF test evaluates whether the log expenditure series Y_t contains a unit root (is non-stationary):

- **H0:** Unit root present → series is non-stationary
- **H1:** No unit root → series is stationary
- **Reject H0** if p-value < 0.05

```
cat("=== ADF Test: Log Government Expenditure (Level) ===\n")
adf_level <- adf.test(Y)
print(adf_level)

if (adf_level$p.value > 0.05) {
  cat("\n>> Decision: Fail to reject H0 (p =",
     round(adf_level$p.value, 4),
  ")     Series is NON-STATIONARY. Differencing required.\n")
} else {
  cat("\n>> Decision: Reject H0 (p =",
     round(adf_level$p.value, 4),
  ")     Series is STATIONARY. No differencing needed.\n")
}
```

```
=== ADF Test: Log Government Expenditure (Level) ===

Augmented Dickey-Fuller Test

data:  Y
Dickey-Fuller = -2.2832, Lag order = 3, p-value = 0.4626
alternative hypothesis: stationary


>> Decision: Fail to reject H0 (p = 0.4626 ) - Series is NON-STATIONARY. Differencing required.
```

4.5.2 Second-Order Differencing

If the level series is non-stationary, differencing is applied to remove the trend and induce stationarity. Second-order differencing (d = 2) corresponds to the integration order I(2),

which is common for fiscal series with strong exponential growth.

```
# Apply second-order differencing
log_y_diff <- diff(Y, differences = 2)

cat("=== ADF Test: Second-Order Differenced Series ===\n")
adf_diff <- adf.test(log_y_diff)
print(adf_diff)

if (adf_diff$p.value < 0.05) {
        cat("\n>> Decision: Reject H0 (p =", round(adf_diff$p.value, 4),
        ") - Differenced series is STATIONARY. d = 2 is appropriate.\n")
} else {
        cat("\n>> Decision: Fail to reject H0 - further differencing may be required.\n")
}
```

```
=== ADF Test: Second-Order Differenced Series ===
Warning in adf.test(log_y_diff) : p-value smaller than
   printed p-value
Augmented Dickey-Fuller Test

data:  log_y_diff
Dickey-Fuller = -4.8497, Lag order = 3, p-value = 0.01
alternative hypothesis: stationary


>> Decision: Reject H0 (p = 0.01 )     Differenced series is
   STATIONARY. d = 2 is appropriate.
```

```
#>
#> >> Decision: Reject H0 (p = 0.01 ) --- Differenced series is STATIONARY. d = 2 is appropriate.
```

4.5.3 Visual Comparison: Level vs Differenced Series

```
Log Government Expenditure"}
par(mfrow = c(2, 1), mar = c(4, 5, 3, 2))

plot(Y, type = "o", col = "steelblue", lwd = 1.5,
main = "Level Series: Log Government Expenditure",
ylab = "Log(USD)", xlab = "")
abline(lm(Y ~ time(Y)), col = "red", lty = 2, lwd = 2)
legend("topleft", legend = "OLS Trend Line", col = "red",
```

```
lty = 2, bty = "n", cex = 0.85)
grid(col = "lightgray", lty = "dotted")

plot(log_y_diff, type = "o", col = "firebrick", lwd = 1.5,
main = "Second-Order Differenced Series (d = 2)",
ylab = "Differenced Log(USD)", xlab = "Year")
abline(h = 0, col = "black", lty = 2)
grid(col = "lightgray", lty = "dotted")
```

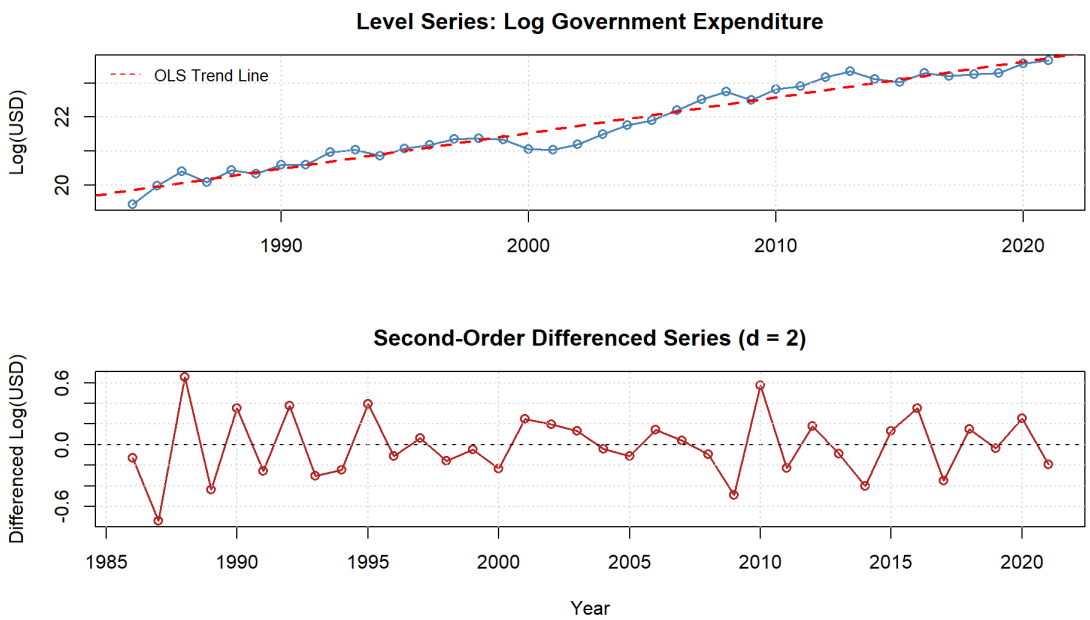

Figure 3: Level and Second-Differenced Log Government Expenditure

Interpretation: The level series (top panel) shows a clear upward trend with a fitted OLS trend line, confirming non-stationarity. After second-order differencing (bottom panel), the series fluctuates around zero without a systematic drift, indicating that the trend has been successfully removed. The ADF test result on the differenced series confirms stationarity, and d = 2 is adopted in the ARIMA/ARIMAX specifications.

4.6 ACF and PACF Analysis — Identifying Model Orders

The **Autocorrelation Function (ACF)** and **Partial Autocorrelation Function (PACF)** are the primary tools for identifying the autoregressive order (p) and moving average order (q) in the Box-Jenkins framework:

Pattern	Implication
PACF: spike at lag k, then cuts off	AR(k) component
ACF: spike at lag k, then cuts off	MA(k) component
Both decay slowly	Mixed ARMA; use `auto.arima`
No significant spikes in differenced series	Differencing sufficient; d is appropriate

```
par(mfrow = c(1, 2), mar = c(4, 4, 3, 1))
Acf(Y,  main = "ACF     Level Series",  lag.max = 20); grid()
Pacf(Y, main = "PACF     Level Series", lag.max = 20); grid()
```

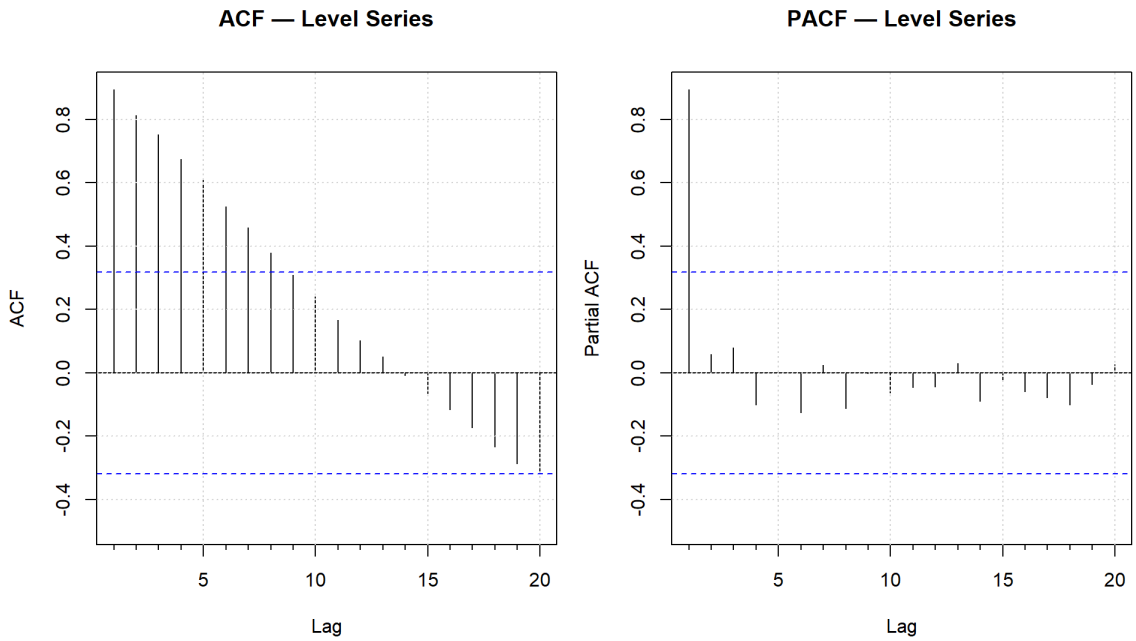

Figure 4: ACF and PACF of Level Series (Log Government Expenditure)

```
par(mfrow = c(1, 2), mar = c(4, 4, 3, 1))
Acf(log_y_diff,  main = "ACF     Differenced Series",  lag.max
   = 20); grid()
Pacf(log_y_diff, main = "PACF     Differenced Series", lag.max
   = 20); grid()
```

Interpretation: The level series ACF decays slowly (consistent with non-stationarity). In the differenced series, spikes that fall within the 95% confidence bands (blue dashed lines at $\pm 1.96/\sqrt{n}$) suggest that differencing has removed most autocorrelation. Remaining significant spikes, if any, guide the

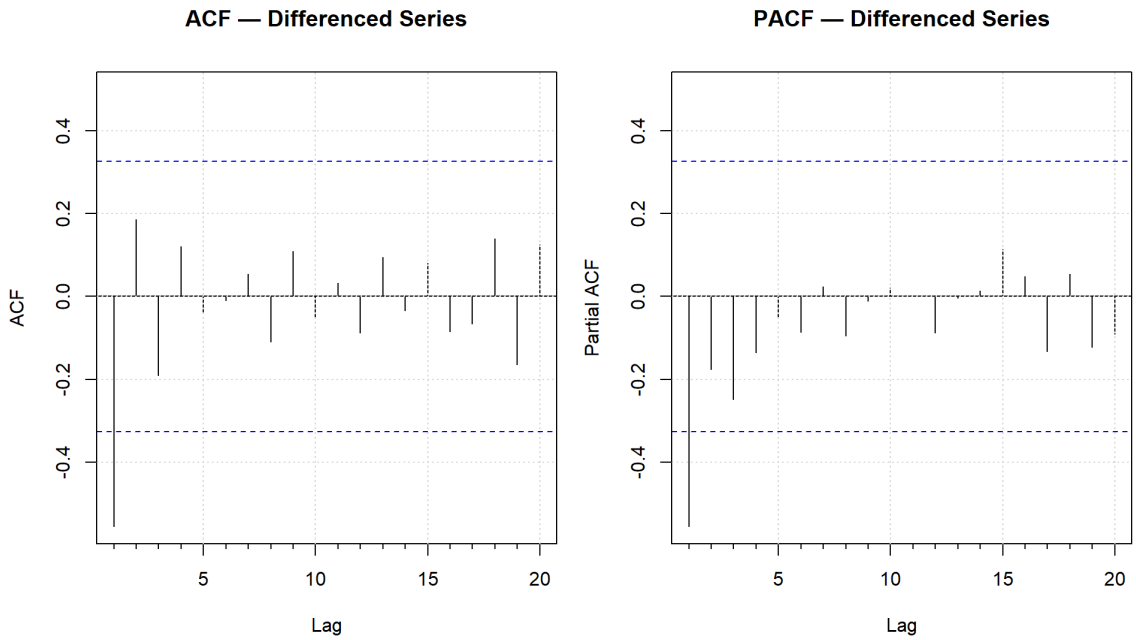

Figure 5: ACF and PACF of Second-Differenced Series

selection of p and q. `auto.arima()` in subsequent sections performs a systematic AIC/BIC-guided search to confirm the optimal orders.

4.7 Defining Forecast Variables

Future values of the exogenous variables must be specified to generate ARIMAX out-of-sample forecasts. A **mean-projection approach** is used here, setting each exogenous variable to its in-sample mean across the 10-year horizon. This is a conservative, assumption-neutral approach.

In practice, scenario-based projections (e.g., IMF WEO GDP projections, UN population forecasts) would be preferred for policy applications. The mean projection provides a baseline forecast that captures the average historical influence of each exogenous variable.

```
# Mean-based projection of exogenous variables over 10-year
  horizon
future_population <- rep(mean(data$POPULATION), h_period)
future_revenue    <- rep(mean(data$LOG_REVENUE), h_period)
future_gdp        <- rep(mean(data$LOG_GDP),     h_period)
```

```
# Assemble as a matrix with column names matching the
   estimation xreg
future_xreg <- cbind(future_population, future_revenue,
   future_gdp)
colnames(future_xreg) <- colnames(xreg_vars)

cat("Future exogenous variable projections (mean of in-sample
   values):\n")
cat("Population (mean):", round(mean(data$POPULATION), 0),
   "\n")
cat("Log Revenue (mean):", round(mean(data$LOG_REVENUE), 4),
   "\n")
cat("Log GDP (mean):", round(mean(data$LOG_GDP), 4), "\n")
```

```
Future exogenous variable projections (mean of in-sample values):
Population (mean): 21768557
Log Revenue (mean): 21.5995
Log GDP (mean): 23.3838
```

4.8 Model 1: ARIMA Baseline

The univariate ARIMA model serves as the benchmark against which all ARIMAX specifications are evaluated. It explains Ghana's log expenditure solely from its own historical dynamics — past values and lagged shocks — with no external information.

4.8.1 Model Estimation

`auto.arima()` with `stepwise = FALSE` and `approximation = FALSE` performs an exhaustive search across candidate (p, d, q) combinations, selecting the specification with the lowest AIC. Maximum Likelihood Estimation (MLE) is used for parameter estimation, providing efficient and asymptotically unbiased estimates.

```
cat("=== Fitting ARIMA Baseline Model ===\n")
cat("Method: Maximum Likelihood Estimation (MLE)\n")
cat("Selection: AIC-minimising exhaustive search\n\n")
```

```r
arima_fit <- auto.arima(
Y,
stepwise       = FALSE,    # Exhaustive, not stepwise, search
approximation  = FALSE,    # Exact (not approximate) AIC values
method         = "ML"      # Maximum Likelihood Estimation
)

summary(arima_fit)
```

```
=== Fitting ARIMA Baseline Model ===
Method: Maximum Likelihood Estimation (MLE)
Selection: AIC-minimising exhaustive search

Series: Y
ARIMA(0,1,0) with drift

Coefficients:
drift
0.1142
s.e.  0.0340

sigma^2 = 0.0441:  log likelihood = 5.76
AIC=-7.51   AICc=-7.16   BIC=-4.29

Training set error measures:
ME      RMSE      MAE         MPE      MAPE      MASE        ACF1
Training set 0.0005083704 0.2043941 0.168418 0.004295336 0.7818951 0.8428475 -0.09354056
```

4.8.2 Residual Diagnostics

A valid ARIMA model should produce **white noise residuals** — residuals that are uncorrelated, approximately normally distributed, and have constant variance. Any remaining structure in the residuals indicates model misspecification.

```r
checkresiduals(arima_fit)
```

4.8.3 Ljung-Box Test for Residual Autocorrelation

The Ljung-Box test formally tests whether the first 10 residual autocorrelations are jointly zero:

- **H0:** Residuals are uncorrelated (white noise) — model is adequate

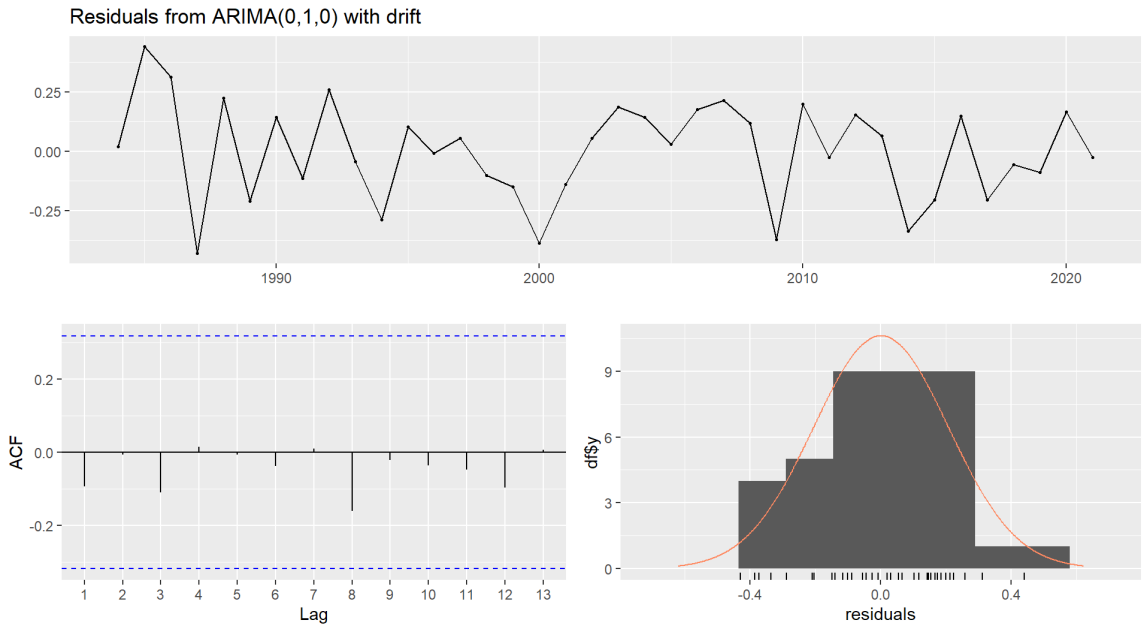

Figure 6: ARIMA Baseline — Residual Diagnostics

- **H1:** Significant autocorrelation remains — model may be misspecified
- **Rule:** p-value $> 0.05 \rightarrow$ fail to reject H0 $\rightarrow$ residuals are clean

```
lb_arima <- Box.test(residuals(arima_fit), lag = 10, type =
   "Ljung-Box")
cat("=== Ljung-Box Test (lag = 10)     ARIMA Baseline ===\n")
print(lb_arima)

if (lb_arima$p.value > 0.05) {
  cat("\n>> Decision: p =", round(lb_arima$p.value, 4),
  "> 0.05     Fail to reject H0. Residuals are white noise.
     Model is adequate.\n")
} else {
  cat("\n>> Decision: p =", round(lb_arima$p.value, 4),
  "< 0.05     Reject H0. Residual autocorrelation detected.
     Model may be misspecified.\n")
}
```

```
=== Ljung-Box Test (lag = 10) - ARIMA Baseline ===

Box-Ljung test

data:  residuals(arima_fit)
```

```
X-squared = 2.3896, df = 10, p-value = 0.9924


>> Decision: p = 0.9924 > 0.05 - Fail to reject H0. Residuals are white noise. Model is adequate.
```

4.8.4 Information Criteria and Accuracy

```r
cat("=== ARIMA     Model Fit Statistics ===\n")
cat("AIC:", round(AIC(arima_fit), 4), "\n")
cat("BIC:", round(BIC(arima_fit), 4), "\n\n")
cat("=== ARIMA     In-Sample Accuracy Metrics ===\n")
print(accuracy(arima_fit))
```

```
=== ARIMA - Model Fit Statistics ===
AIC: -7.5107
BIC: -4.2889

=== ARIMA - In-Sample Accuracy Metrics ===
ME      RMSE      MAE         MPE      MAPE      MASE        ACF1
Training set 0.0005083704 0.2043941 0.168418 0.004295336 0.7818951 0.8428475 -0.09354056
```

4.8.5 ARIMA Forecast (10-Year Horizon)

```r
arima_forecast <- forecast(arima_fit, h = h_period)

cat("=== ARIMA     Point Forecasts and Prediction Intervals
   ===\n")
print(arima_forecast)

par(mfrow = c(1, 1), mar = c(4, 5, 3, 2))
plot(arima_forecast,
main = "ARIMA Baseline: 10-Year Forecast     Log Government
   Expenditure",
ylab = "Log Government Expenditure (USD)",
xlab = "Year",
col  = "steelblue")
grid(col = "lightgray", lty = "dotted")
```

```
=== ARIMA - Point Forecasts and Prediction Intervals ===
Point Forecast    Lo 80    Hi 80    Lo 95    Hi 95
```

```
2022        23.77176 23.50264 24.04088 23.36018 24.18334
2023        23.88595 23.50536 24.26655 23.30389 24.46802
2024        24.00015 23.53402 24.46628 23.28727 24.71303
2025        24.11435 23.57611 24.65259 23.29118 24.93751
2026        24.22854 23.62678 24.83031 23.30822 25.14887
2027        24.34274 23.68354 25.00195 23.33457 25.35091
2028        24.45694 23.74492 25.16896 23.36799 25.54588
2029        24.57113 23.80995 25.33232 23.40700 25.73527
2030        24.68533 23.87797 25.49269 23.45058 25.92008
2031        24.79953 23.94850 25.65056 23.49799 26.10107
```

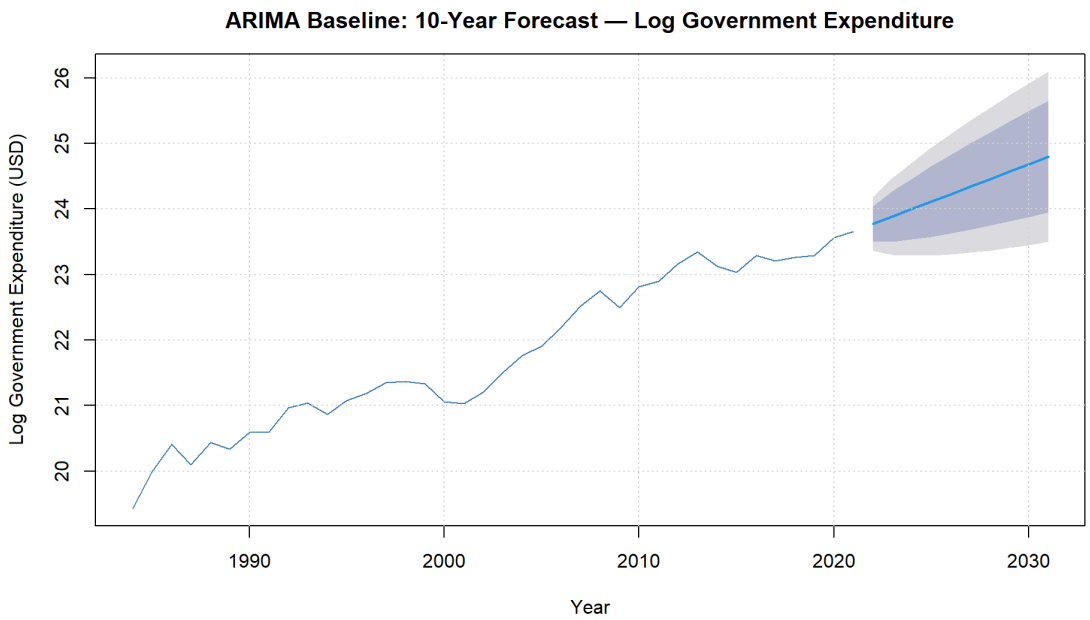

Figure 7: ARIMA — 10-Year Forecast of Log Government Expenditure

---

4.9 Model 2: ARIMAX — All Exogenous Variables

This ARIMAX model includes all three exogenous variables simultaneously: population, log revenue, and log GDP. It tests whether the combined macroeconomic signal from these variables improves on the univariate ARIMA baseline.

```
cat("=== Fitting ARIMAX     All Exogenous Variables ===\n")
cat("Exogenous variables: POPULATION + LOG_REVENUE +
   LOG_GDP\n\n")

arimax_fit <- auto.arima(
Y,
```

```
xreg           = as.matrix(xreg_vars),
stepwise       = FALSE,
approximation  = FALSE,
method         = "ML"
)

summary(arimax_fit)
```

```
=== Fitting ARIMAX - All Exogenous Variables ===
Exogenous variables: POPULATION + LOG_REVENUE + LOG_GDP

Series: Y
Regression with ARIMA(0,0,0) errors

Coefficients:
intercept  POPULATION  LOG_REVENUE  LOG_GDP
2.2976      0.0000       0.8356    0.0247
s.e.     1.3758      0.0002       0.0758    0.0712

sigma^2 = 0.01478:  log likelihood = 28.26
AIC=-46.53   AICc=-44.65   BIC=-38.34

Training set error measures:
ME      RMSE        MAE          MPE      MAPE      MASE      ACF1
Training set -0.00000000000000000934926 0.1150152 0.09975323 -0.003032057 0.4592601 0.4992147 0.2935644
```

```
checkresiduals(arimax_fit)
```

```
lb_all <- Box.test(residuals(arimax_fit), lag = 10, type = "Ljung-Box")
cat("=== Ljung-Box Test - ARIMAX (All Variables) ===\n")
print(lb_all)
cat("\n>> Decision: p =", round(lb_all$p.value, 4), "-",
ifelse(lb_all$p.value > 0.05,
"Fail to reject H0. Residuals are white noise.",
"Reject H0. Residual autocorrelation present."), "\n")
```

```
=== Ljung-Box Test      ARIMAX (All Variables) ===

Box-Ljung test

data:  residuals(arimax_fit)
X-squared = 10.955, df = 10, p-value = 0.3611
```

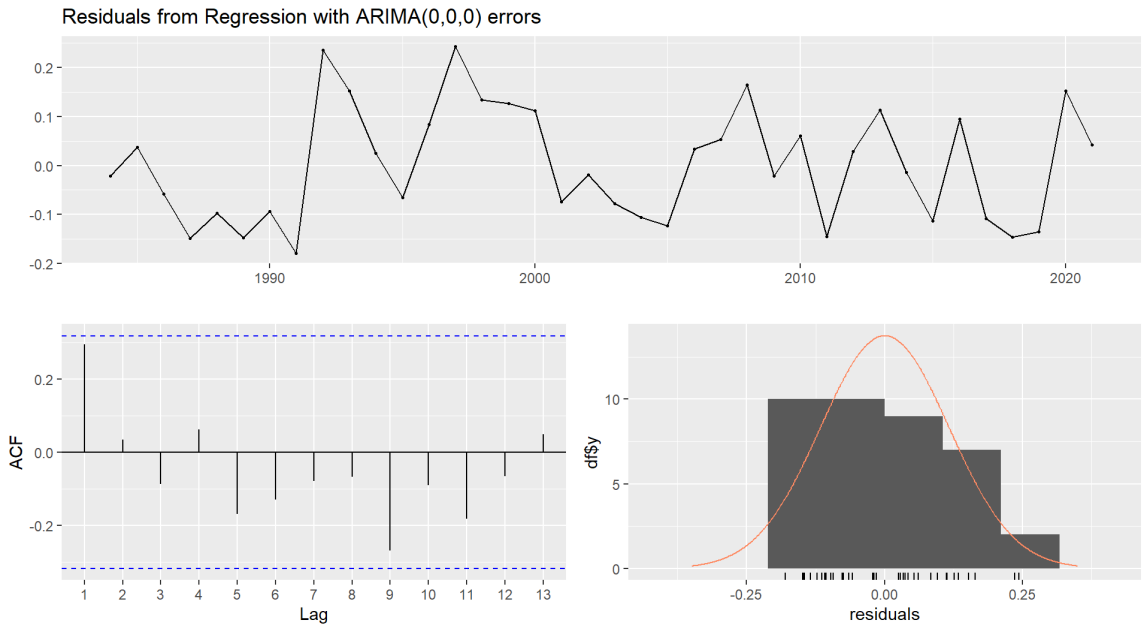

Figure 8: ARIMAX (All Variables) — Residual Diagnostics

```
>> Decision: p = 0.3611     Fail to reject H0. Residuals are
   white noise.
```

```
cat("=== ARIMAX (All Variables)     Fit Statistics ===\n")
cat("AIC:", round(AIC(arimax_fit), 4), "\n")
cat("BIC:", round(BIC(arimax_fit), 4), "\n\n")
cat("=== ARIMAX (All Variables)     Accuracy Metrics ===\n")
print(accuracy(arimax_fit))
```

```
=== ARIMAX (All Variables) - Fit Statistics ===
AIC: -46.5252
BIC: -38.3372

=== ARIMAX (All Variables) - Accuracy Metrics ===
ME      RMSE        MAE          MPE      MAPE      MASE      ACF1
Training set -0.000000000000000934926 0.1150152 0.09975323 -0.003032057 0.4592601 0.4992147 0.2935644
```

```
arimax_all_forecast <- forecast(arimax_fit, xreg =
   future_xreg, h = h_period)

cat("=== ARIMAX (All Variables)     Point Forecasts ===\n")
print(arimax_all_forecast)

par(mfrow = c(1, 1))
```

```r
plot(arimax_all_forecast,
main = "ARIMAX (Population + Revenue + GDP): 10-Year Forecast",
ylab = "Log Government Expenditure (USD)",
xlab = "Year")
grid(nx = 30, ny = 30, col = "lightgray", lty = "dotted")
```

```
=== ARIMAX (All Variables) - Point Forecasts ===
Point Forecast    Lo 80    Hi 80    Lo 95    Hi 95
2022       21.79411 21.63828 21.94994 21.55579 22.03243
2023       21.79411 21.63828 21.94994 21.55579 22.03243
2024       21.79411 21.63828 21.94994 21.55579 22.03243
2025       21.79411 21.63828 21.94994 21.55579 22.03243
2026       21.79411 21.63828 21.94994 21.55579 22.03243
2027       21.79411 21.63828 21.94994 21.55579 22.03243
2028       21.79411 21.63828 21.94994 21.55579 22.03243
2029       21.79411 21.63828 21.94994 21.55579 22.03243
2030       21.79411 21.63828 21.94994 21.55579 22.03243
2031       21.79411 21.63828 21.94994 21.55579 22.03243
```

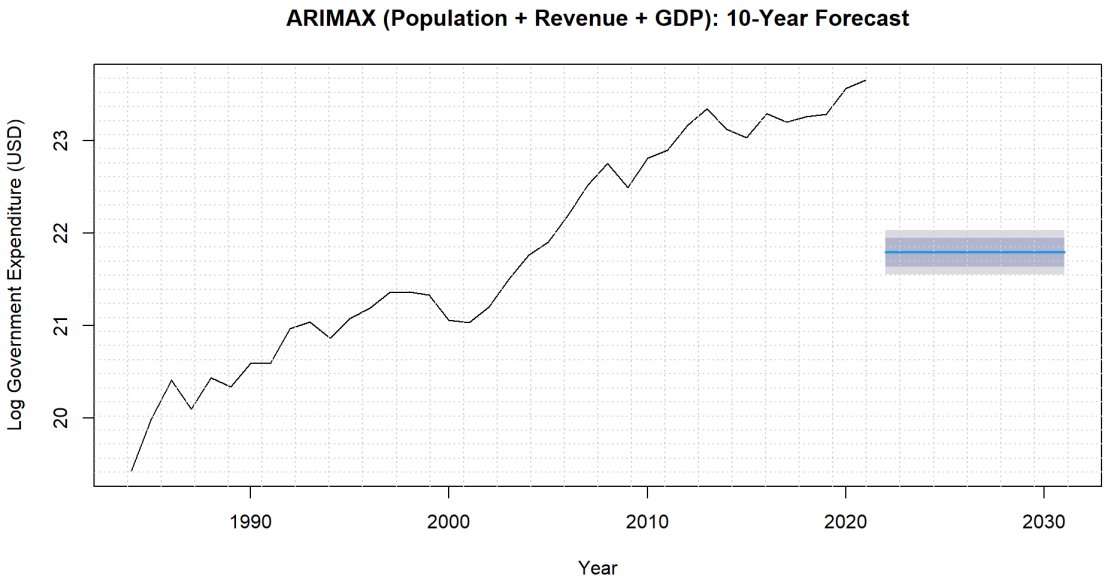

Figure 9: ARIMAX (All Variables) — 10-Year Forecast

4.9.1 Pairwise Comparison: ARIMA vs ARIMAX (All)

```r
comp_all <- data.frame(
      Metric      = c("AIC", "BIC", "ME", "RMSE", "MAE", "MPE",
         "MAPE", "MASE"),
```

```
      ARIMA       = c(AIC(arima_fit), BIC(arima_fit),
  accuracy(arima_fit)[1, "ME"],    accuracy(arima_fit)[1,
     "RMSE"],
  accuracy(arima_fit)[1, "MAE"],   accuracy(arima_fit)[1,
     "MPE"],
  accuracy(arima_fit)[1, "MAPE"], accuracy(arima_fit)[1,
     "MASE"]),
      ARIMAX_ALL = c(AIC(arimax_fit), BIC(arimax_fit),
  accuracy(arimax_fit)[1, "ME"],    accuracy(arimax_fit)[1,
     "RMSE"],
  accuracy(arimax_fit)[1, "MAE"],   accuracy(arimax_fit)[1,
     "MPE"],
  accuracy(arimax_fit)[1, "MAPE"], accuracy(arimax_fit)[1,
     "MASE"])
) |>
mutate(
  Improvement = ifelse(Metric %in% c("ME", "MPE"),
    ifelse(abs(ARIMAX_ALL) < abs(ARIMA), "Yes", "No"),
      ifelse(ARIMAX_ALL < ARIMA, "Yes", "No"))
)

knitr::kable(comp_all, digits = 4,
caption = "Table 2: ARIMA vs ARIMAX (All Variables)")
```

Table 4: Table 2: ARIMA vs ARIMAX (All Variables)

Metric	ARIMA	ARIMAX_ALL	Improvement
AIC	-7.5107	-46.5252	Yes
BIC	-4.2889	-38.3372	Yes
ME	0.0005	0.0000	Yes
RMSE	0.2044	0.1150	Yes
MAE	0.1684	0.0998	Yes
MPE	0.0043	-0.0030	Yes
MAPE	0.7819	0.4593	Yes
MASE	0.8428	0.4992	Yes

4.10 Model 3: ARIMAX — Population Only

Population is included as a standalone exogenous variable. In line with Wagner's Law, population growth increases aggregate demand for public services — education, healthcare, infrastructure — directly pressuring expenditure upward. This model isolates the demographic channel.

```
cat("=== Fitting ARIMAX     Population as Exogenous Variable
   ===\n\n")

arimax_fit_pop <- auto.arima(
Y,
xreg           = as.matrix(xreg_vars[, "POPULATION"]),
stepwise       = FALSE,
approximation  = FALSE,
method         = "ML"
)

summary(arimax_fit_pop)
```

```
=== Fitting ARIMAX - Population as Exogenous Variable ===

Series: Y
Regression with ARIMA(0,0,0) errors

Coefficients:
intercept  POPULATION
17.5005      0.0000
s.e.     0.1676      0.0002

sigma^2 = 0.07521:  log likelihood = -3.73
AIC=13.46   AICc=14.17   BIC=18.38

Training set error measures:
ME      RMSE       MAE         MPE     MAPE     MASE      ACF1
Training set 0.0000000000001058335 0.2669384 0.2195247 -0.01651598 1.008788 1.098611 0.5917734
```

```
  checkresiduals(arimax_fit_pop)
```

```
lb_pop <- Box.test(residuals(arimax_fit_pop), lag = 10, type = "Ljung-Box")
cat("=== Ljung-Box Test - ARIMAX (Population) ===\n")
print(lb_pop)
cat("\n>> Decision: p =", round(lb_pop$p.value, 4), "-",
ifelse(lb_pop$p.value > 0.05, "Residuals are white noise.", "Autocorrelation detected."), "\n")
```

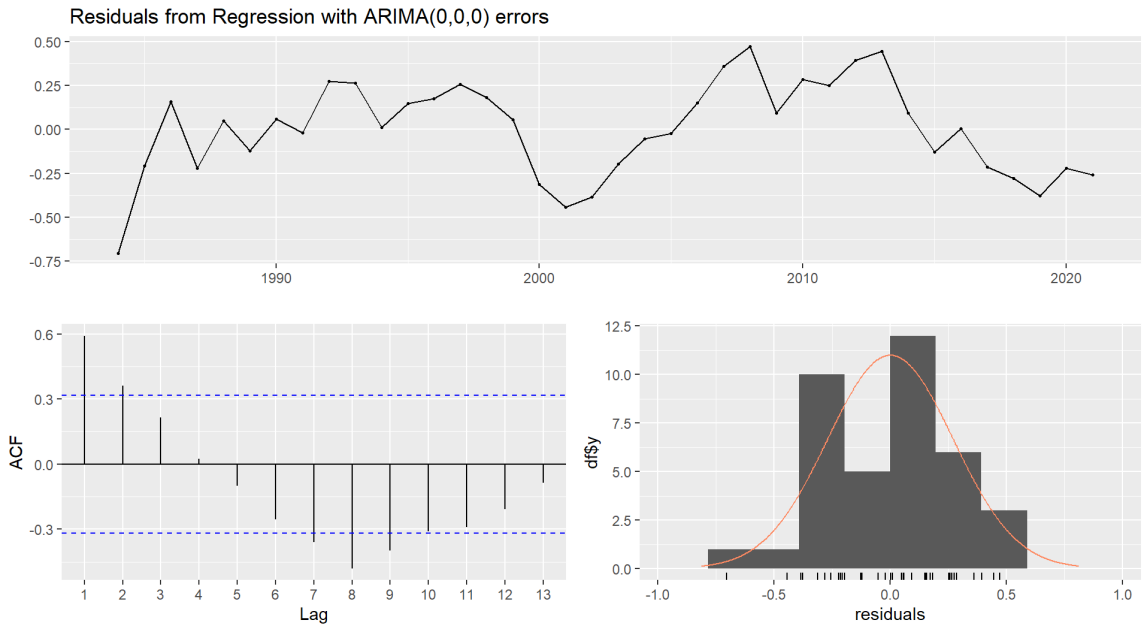

Figure 10: ARIMAX (Population Only) — Residual Diagnostics

```
=== Ljung-Box Test      ARIMAX (Population) ===

Box-Ljung test

data:  residuals(arimax_fit_pop)
X-squared = 57.199, df = 10, p-value = 0.00000001223


>> Decision: p = 0      Autocorrelation detected.
```

```
cat("AIC:", round(AIC(arimax_fit_pop), 4), "  BIC:",
   round(BIC(arimax_fit_pop), 4), "\n")
print(accuracy(arimax_fit_pop))
```

```
AIC: 13.4633   BIC: 18.376
ME      RMSE       MAE         MPE     MAPE     MASE      ACF1
Training set 0.0000000000001058335 0.2669384 0.2195247 -0.01651598 1.008788 1.098611 0.5917734
```

```
pop_forecast <- forecast(arimax_fit_pop,
  xreg = future_xreg[, "POPULATION"],
  h = h_period)
print(pop_forecast)
```

```
plot(pop_forecast,
  main = "ARIMAX (Population Only): 10-Year Forecast",
  ylab = "Log Government Expenditure (USD)", xlab = "Year")
grid(col = "lightgray", lty = "dotted")
```

```
     Point Forecast    Lo 80    Hi 80    Lo 95    Hi 95
2022       21.79411 21.44264 22.14558 21.25658 22.33163
2023       21.79411 21.44264 22.14558 21.25658 22.33163
2024       21.79411 21.44264 22.14558 21.25658 22.33163
2025       21.79411 21.44264 22.14558 21.25658 22.33163
2026       21.79411 21.44264 22.14558 21.25658 22.33163
2027       21.79411 21.44264 22.14558 21.25658 22.33163
2028       21.79411 21.44264 22.14558 21.25658 22.33163
2029       21.79411 21.44264 22.14558 21.25658 22.33163
2030       21.79411 21.44264 22.14558 21.25658 22.33163
2031       21.79411 21.44264 22.14558 21.25658 22.33163
```

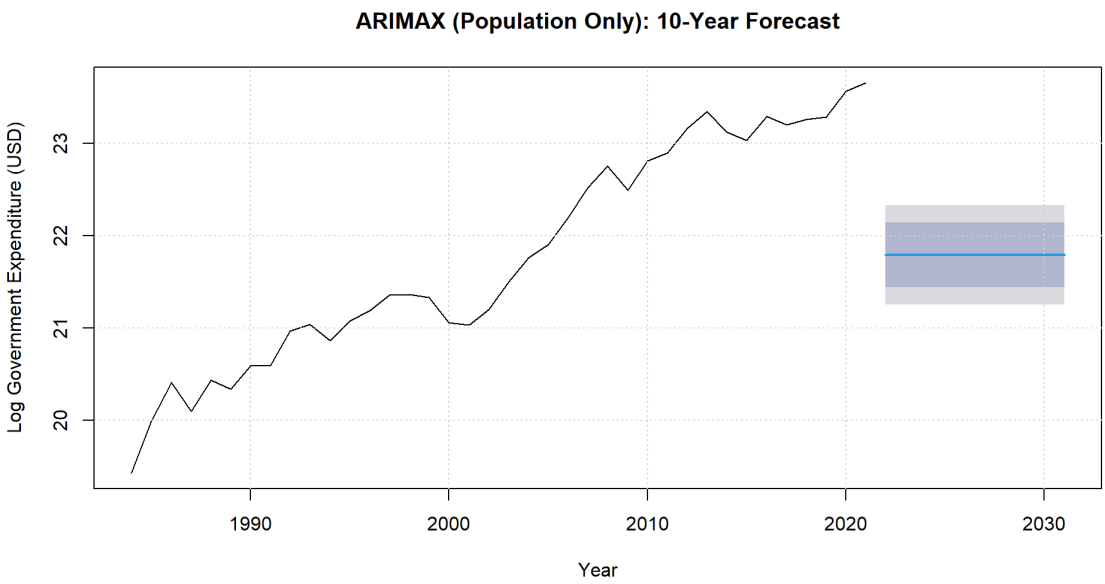

Figure 11: ARIMAX (Population Only) — 10-Year Forecast

4.10.1 Pairwise Comparison: ARIMA vs ARIMAX (Population)

```
comp_rev <- data.frame(
  Metric      = c("AIC", "BIC", "ME", "RMSE", "MAE", "MPE",
     "MAPE", "MASE"),
  ARIMA       = c(AIC(arima_fit), BIC(arima_fit),
```

```
accuracy(arima_fit)[1, "ME"],        accuracy(arima_fit)[1,
   "RMSE"],
accuracy(arima_fit)[1, "MAE"],       accuracy(arima_fit)[1,
   "MPE"],
accuracy(arima_fit)[1, "MAPE"],      accuracy(arima_fit)[1,
   "MASE"]),
  ARIMAX_REV = c(AIC(arimax_fit_rev), BIC(arimax_fit_rev),
accuracy(arimax_fit_rev)[1, "ME"],
   accuracy(arimax_fit_rev)[1, "RMSE"],
accuracy(arimax_fit_rev)[1, "MAE"],
   accuracy(arimax_fit_rev)[1, "MPE"],
accuracy(arimax_fit_rev)[1, "MAPE"],
   accuracy(arimax_fit_rev)[1, "MASE"])
) |>
mutate(
  Improvement = ifelse(Metric %in% c("ME", "MPE"),
    ifelse(abs(ARIMAX_REV) < abs(ARIMA), "Yes", "No"),
      ifelse(ARIMAX_REV < ARIMA, "Yes", "No"))
)

knitr::kable(comp_rev, digits = 4,
caption = "Table 4: ARIMA vs ARIMAX (Revenue Only)")
```

Table 5: Table 3: ARIMA vs ARIMAX (Population Only)

Metric	ARIMA	ARIMAX_POP	Improvement
AIC	-7.5107	13.4633	No
BIC	-4.2889	18.3760	No
ME	0.0005	0.0000	Yes
RMSE	0.2044	0.2669	No
MAE	0.1684	0.2195	No
MPE	0.0043	-0.0165	No
MAPE	0.7819	1.0088	No
MASE	0.8428	1.0986	No

4.11 Model 4: ARIMAX — Revenue Only

Government revenue is a direct fiscal variable and a primary determinant of spending capacity. Ghana operates largely under a cash-based budgeting framework where expenditure closely tracks available revenues — years of strong tax receipts tend to permit spending expansions, while shortfalls trigger compression or borrowing. This model isolates the revenue-fiscal transmission channel.

```
cat("=== Fitting ARIMAX     Log Revenue as Exogenous Variable
   ===\n\n")

arimax_fit_rev <- auto.arima(
Y,
xreg           = as.matrix(xreg_vars[, "LOG_REVENUE"]),
stepwise       = FALSE,
approximation  = FALSE,
method         = "ML"
)

summary(arimax_fit_rev)
```

```
=== Fitting ARIMAX - Log Revenue as Exogenous Variable ===

Series: Y
Regression with ARIMA(1,0,0) errors

Coefficients:
ar1  LOG_REVENUE
0.4729       1.0094
s.e.  0.1453       0.0018

sigma^2 = 0.01693:  log likelihood = 24.47
AIC=-42.95   AICc=-42.24   BIC=-38.04

Training set error measures:
ME      RMSE       MAE         MPE      MAPE      MASE        ACF1
Training set -0.001601658 0.1266484 0.1022711 -0.01889309 0.4701961 0.5118156 0.005738759
```

```
checkresiduals(arimax_fit_rev)
```

```
lb_rev <{-} Box.test(residuals(arimax_fit_rev), lag = 10, type
   = "Ljung{-}Box")
cat("=== Ljung{-}Box Test --- ARIMAX (Revenue) ===={}n")
```

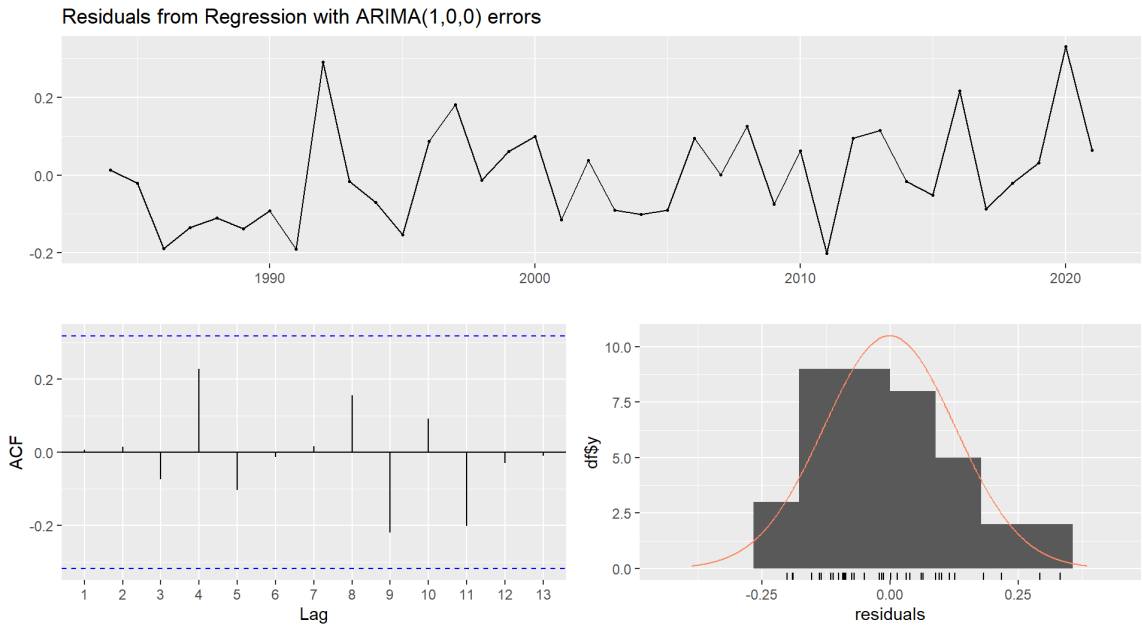

Figure 12: ARIMAX (Revenue Only) — Residual Diagnostics

```
print(lb_rev)
```

```
cat("AIC:", round(AIC(arimax_fit_rev), 4), "  BIC:",
   round(BIC(arimax_fit_rev), 4), "{}n")

print(accuracy(arimax_fit_rev))
```

```
  AIC: -42.9495   BIC: -38.0367
                    ME      RMSE       MAE         MPE      MAPE      MASE
Training set -0.001601658 0.1266484 0.1022711 -0.01889309 0.4701961 0.5118156
                   ACF1
Training set 0.005738759
```

```
rev_forecast <- forecast(arimax_fit_rev,
xreg = future_xreg[, "LOG_REVENUE"],
h    = h_period)
print(rev_forecast)

plot(rev_forecast,
main = "ARIMAX (Revenue Only): 10-Year Forecast",
ylab = "Log Government Expenditure (USD)", xlab = "Year")
grid(col = "lightgray", lty = "dotted")
```

```
     Point Forecast    Lo 80    Hi 80    Lo 95    Hi 95
2022       21.90921 21.74246 22.07596 21.65418 22.16424
2023       21.85290 21.66844 22.03736 21.57080 22.13500
2024       21.82627 21.63809 22.01446 21.53846 22.11408
2025       21.81368 21.62467 22.00269 21.52461 22.10275
2026       21.80773 21.61853 21.99692 21.51838 22.09708
2027       21.80491 21.61568 21.99415 21.51550 22.09433
2028       21.80358 21.61434 21.99283 21.51416 22.09301
2029       21.80295 21.61370 21.99220 21.51352 22.09238
2030       21.80265 21.61341 21.99190 21.51322 22.09208
2031       21.80251 21.61327 21.99176 21.51308 22.09194
```

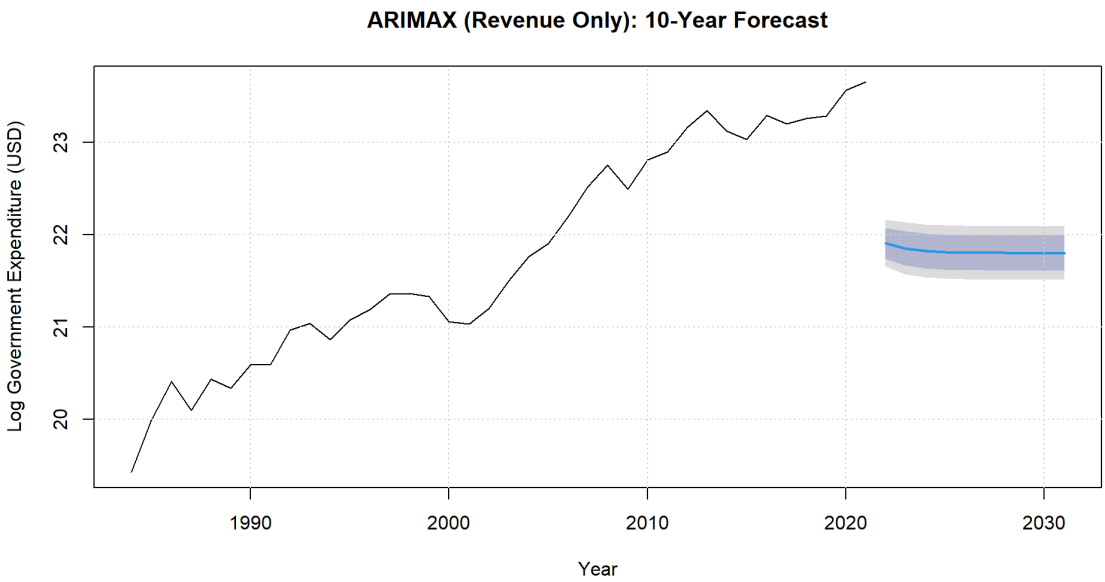

Figure 13: ARIMAX (Revenue Only) — 10-Year Forecast

4.11.1 Pairwise Comparison: ARIMA vs ARIMAX (Revenue)

```r
comp_rev <- data.frame(
Metric      = c("AIC", "BIC", "ME", "RMSE", "MAE", "MPE",
   "MAPE", "MASE"),
ARIMA       = c(AIC(arima_fit), BIC(arima_fit),
accuracy(arima_fit)[1, "ME"],        accuracy(arima_fit)[1,
   "RMSE"],
accuracy(arima_fit)[1, "MAE"],       accuracy(arima_fit)[1,
   "MPE"],
accuracy(arima_fit)[1, "MAPE"],      accuracy(arima_fit)[1,
   "MASE"]),
```

```
ARIMAX_REV = c(AIC(arimax_fit_rev), BIC(arimax_fit_rev),
accuracy(arimax_fit_rev)[1, "ME"],
   accuracy(arimax_fit_rev)[1, "RMSE"],
accuracy(arimax_fit_rev)[1, "MAE"],
   accuracy(arimax_fit_rev)[1, "MPE"],
accuracy(arimax_fit_rev)[1, "MAPE"],
   accuracy(arimax_fit_rev)[1, "MASE"])
) |>
mutate(
Improvement = ifelse(Metric %in% c("ME", "MPE"),
ifelse(abs(ARIMAX_REV) < abs(ARIMA), "Yes", "No"),
ifelse(ARIMAX_REV < ARIMA, "Yes", "No"))
)

knitr::kable(comp_rev, digits = 4,
caption = "Table 4: ARIMA vs ARIMAX (Revenue Only)")
```

Table 6: Table 4: ARIMA vs ARIMAX (Revenue Only)

Metric	ARIMA	ARIMAX_REV	Improvement
AIC	-7.5107	-42.9495	Yes
BIC	-4.2889	-38.0367	Yes
ME	0.0005	-0.0016	No
RMSE	0.2044	0.1266	Yes
MAE	0.1684	0.1023	Yes
MPE	0.0043	-0.0189	No
MAPE	0.7819	0.4702	Yes
MASE	0.8428	0.5118	Yes

4.12 Model 5: ARIMAX — GDP Only

GDP captures the overall level of economic activity and is one of the most theoretically motivated exogenous drivers of government expenditure. Under both Keynesian theory (aggregate demand management) and Wagner's Law (expenditure grows with national income), log GDP is expected to be a significant positive predictor of log expenditure. This model isolates the GDP channel.

```
cat("=== Fitting ARIMAX --- Log GDP as Exogenous Variable
   ===\{}n\{}n")
```

```
#> === Fitting ARIMAX --- Log GDP as Exogenous Variable ===
```

```
arimax_fit_gdp <- auto.arima(
Y,
xreg           = as.matrix(xreg_vars[, "LOG_GDP"]),
stepwise       = FALSE,
approximation  = FALSE,
method         = "ML"
)

summary(arimax_fit_gdp)
```

```
=== Fitting ARIMAX - Log GDP as Exogenous Variable ===

Series: Y
Regression with ARIMA(0,1,1) errors

Coefficients:
ma1   drift  LOG_GDP
-0.4116  0.0535   0.6978
s.e.   0.1732  0.0193   0.1350

sigma^2 = 0.02837:  log likelihood = 14.88
AIC=-21.76   AICc=-20.51   BIC=-15.31

Training set error measures:
ME      RMSE       MAE        MPE      MAPE      MASE      ACF1
Training set 0.007469537 0.1593093 0.1270311 0.04706516 0.5882488 0.6357269 0.1125206
```

```
checkresiduals(arimax_fit_gdp)
```

```
lb_gdp <- Box.test(residuals(arimax_fit_gdp), lag = 10, type = "Ljung-Box")
cat("=== Ljung-Box Test - ARIMAX (GDP) ===\n")
print(lb_gdp)
cat("\n>> Decision: p =", round(lb_gdp$p.value, 4), "-",
ifelse(lb_gdp$p.value > 0.05, "Residuals are white noise.", "Autocorrelation detected."), "\n")
```

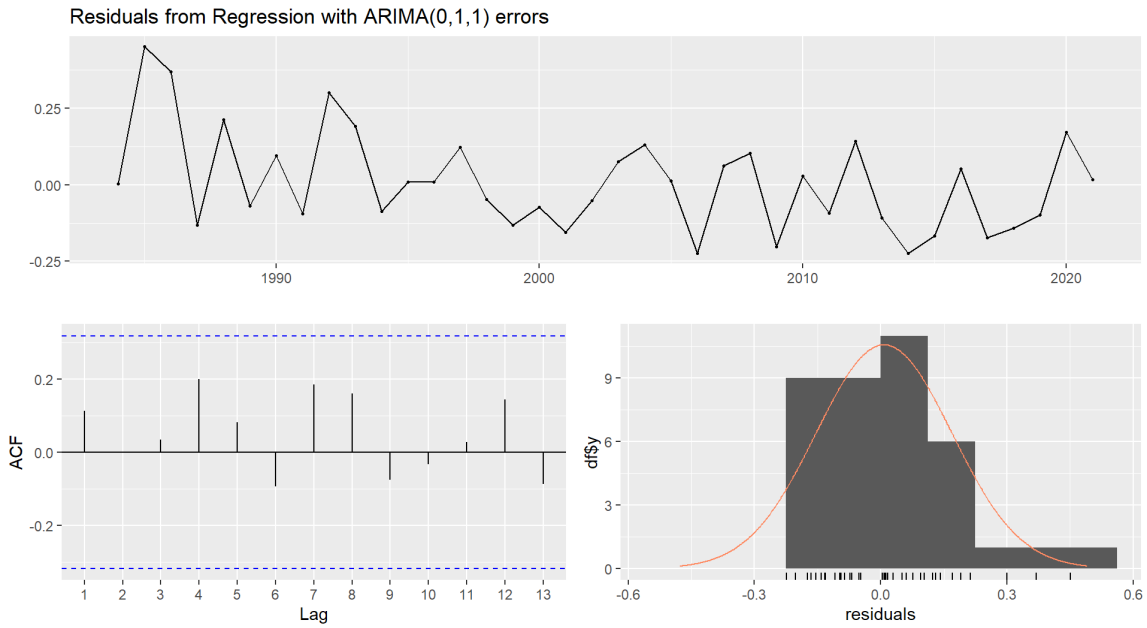

Figure 14: ARIMAX (GDP Only) — Residual Diagnostics

```
=== Ljung-Box Test - ARIMAX (GDP) ===

Box-Ljung test

data:  residuals(arimax_fit_gdp)
X-squared = 6.3895, df = 10, p-value = 0.7815


>> Decision: p = 0.7815 - Residuals are white noise.
```

```
```{r ch4-arimax-gdp-stats}
cat("AIC:", round(AIC(arimax_fit_gdp), 4), "  BIC:",
   round(BIC(arimax_fit_gdp), 4), "\n")
print(accuracy(arimax_fit_gdp))
```
```

```
AIC: -21.758   BIC: -15.3143
ME      RMSE       MAE        MPE      MAPE      MASE      ACF1
Training set 0.007469537 0.1593093 0.1270311 0.04706516 0.5882488 0.6357269 0.1125206
```

```
gdp_forecast <- forecast(arimax_fit_gdp,
xreg = future_xreg[, "LOG_GDP"],
h    = h_period)
print(gdp_forecast)
```

```
plot(gdp_forecast,
main = "ARIMAX (GDP Only): 10-Year Forecast",
ylab = "Log Government Expenditure (USD)", xlab = "Year")
grid(col = "lightgray", lty = "dotted")
```

```
     Point Forecast    Lo 80    Hi 80    Lo 95    Hi 95
2022       22.50728 22.29144 22.72312 22.17718 22.83737
2023       22.56076 22.31033 22.81120 22.17776 22.94377
2024       22.61425 22.33345 22.89505 22.18481 23.04369
2025       22.66773 22.35955 22.97592 22.19641 23.13906
2026       22.72122 22.38789 23.05455 22.21144 23.23100
2027       22.77470 22.41800 23.13141 22.22917 23.32024
2028       22.82819 22.44955 23.20683 22.24911 23.40727
2029       22.88167 22.48230 23.28105 22.27088 23.49246
2030       22.93516 22.51608 23.35424 22.29423 23.57609
2031       22.98864 22.55074 23.42655 22.31893 23.65836
```

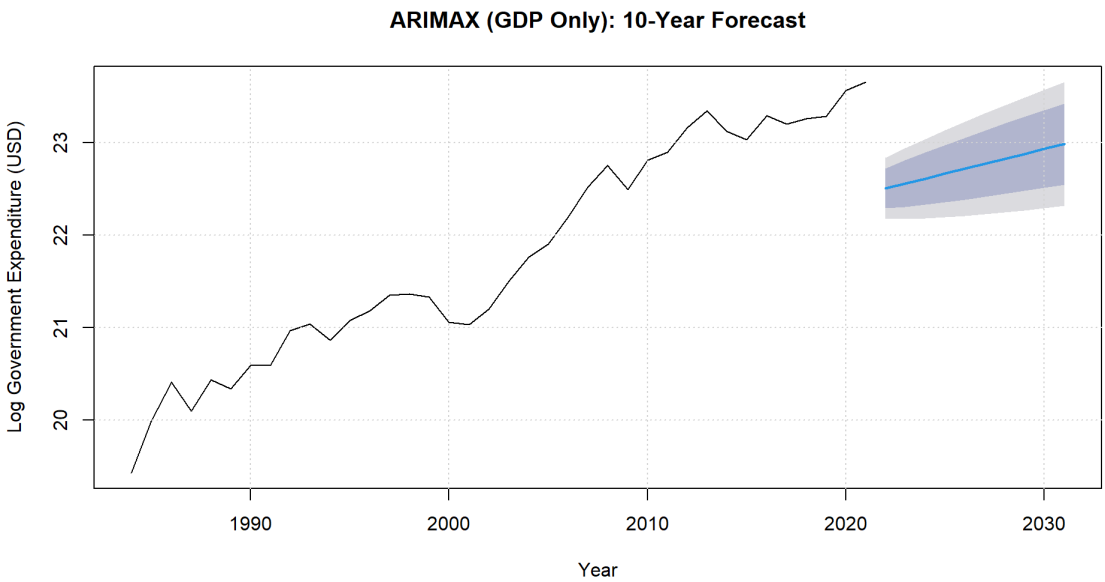

Figure 15: ARIMAX (GDP Only) — 10-Year Forecast

4.12.1 Pairwise Comparison: ARIMA vs ARIMAX (GDP)

```
comp_gdp <- data.frame(
Metric      = c("AIC", "BIC", "ME", "RMSE", "MAE", "MPE",
   "MAPE", "MASE"),
ARIMA       = c(AIC(arima_fit), BIC(arima_fit),
accuracy(arima_fit)[1, "ME"],        accuracy(arima_fit)[1,
   "RMSE"],
```

```
accuracy(arima_fit)[1, "MAE"],      accuracy(arima_fit)[1,
   "MPE"],
accuracy(arima_fit)[1, "MAPE"],     accuracy(arima_fit)[1,
   "MASE"]),
ARIMAX_GDP = c(AIC(arimax_fit_gdp), BIC(arimax_fit_gdp),
accuracy(arimax_fit_gdp)[1, "ME"],
   accuracy(arimax_fit_gdp)[1, "RMSE"],
accuracy(arimax_fit_gdp)[1, "MAE"],
   accuracy(arimax_fit_gdp)[1, "MPE"],
accuracy(arimax_fit_gdp)[1, "MAPE"],
   accuracy(arimax_fit_gdp)[1, "MASE"])
) |>
mutate(
Improvement = ifelse(Metric %in% c("ME", "MPE"),
ifelse(abs(ARIMAX_GDP) < abs(ARIMA), "Yes", "No"),
ifelse(ARIMAX_GDP < ARIMA, "Yes", "No"))
)

knitr::kable(comp_gdp, digits = 4,
caption = "ARIMA vs ARIMAX (GDP Only)")
```

Table 7: ARIMA vs ARIMAX (GDP Only)

Metric	ARIMA	ARIMAX_GDP	Improvement
AIC	-7.5107	-21.7580	Yes
BIC	-4.2889	-15.3143	Yes
ME	0.0005	0.0075	No
RMSE	0.2044	0.1593	Yes
MAE	0.1684	0.1270	Yes
MPE	0.0043	0.0471	No
MAPE	0.7819	0.5882	Yes
MASE	0.8428	0.6357	Yes

4.13 Full Model Comparison — All Five Specifications

This section consolidates all five models into a master comparison table to identify the overall best-performing specification. The **Best_Model** column is determined row-by-row using metric-appropriate criteria:

- For **AIC, BIC, RMSE, MAE, MAPE, MASE**: the model with the **lowest value** wins.
- For **ME and MPE**: the model with the value **closest to zero** (least systematic bias) wins.

```
full_comparison <- data.frame(
Metric = c("AIC", "BIC", "ME", "RMSE", "MAE", "MPE", "MAPE",
   "MASE"),

ARIMA = c(
AIC(arima_fit), BIC(arima_fit),
accuracy(arima_fit)[1, "ME"],   accuracy(arima_fit)[1, "RMSE"],
accuracy(arima_fit)[1, "MAE"],  accuracy(arima_fit)[1, "MPE"],
accuracy(arima_fit)[1, "MAPE"], accuracy(arima_fit)[1, "MASE"]
),
ARIMAX_ALL = c(
AIC(arimax_fit), BIC(arimax_fit),
accuracy(arimax_fit)[1, "ME"],   accuracy(arimax_fit)[1,
   "RMSE"],
accuracy(arimax_fit)[1, "MAE"],  accuracy(arimax_fit)[1,
   "MPE"],
accuracy(arimax_fit)[1, "MAPE"], accuracy(arimax_fit)[1,
   "MASE"]
),
ARIMAX_POP = c(
AIC(arimax_fit_pop), BIC(arimax_fit_pop),
accuracy(arimax_fit_pop)[1, "ME"],
   accuracy(arimax_fit_pop)[1, "RMSE"],
accuracy(arimax_fit_pop)[1, "MAE"],
   accuracy(arimax_fit_pop)[1, "MPE"],
accuracy(arimax_fit_pop)[1, "MAPE"],
   accuracy(arimax_fit_pop)[1, "MASE"]
),
ARIMAX_REV = c(
```

```
AIC(arimax_fit_rev), BIC(arimax_fit_rev),
accuracy(arimax_fit_rev)[1, "ME"],
   accuracy(arimax_fit_rev)[1, "RMSE"],
accuracy(arimax_fit_rev)[1, "MAE"],
   accuracy(arimax_fit_rev)[1, "MPE"],
accuracy(arimax_fit_rev)[1, "MAPE"],
   accuracy(arimax_fit_rev)[1, "MASE"]
),
ARIMAX_GDP = c(
AIC(arimax_fit_gdp), BIC(arimax_fit_gdp),
accuracy(arimax_fit_gdp)[1, "ME"],
   accuracy(arimax_fit_gdp)[1, "RMSE"],
accuracy(arimax_fit_gdp)[1, "MAE"],
   accuracy(arimax_fit_gdp)[1, "MPE"],
accuracy(arimax_fit_gdp)[1, "MAPE"],
   accuracy(arimax_fit_gdp)[1, "MASE"]
))
# Identify best model per metric using correct optimality
   criterion per metric
full_comparison <- full_comparison |>
rowwise() |>
mutate(Best_Model = case_when(
# Lower is better for AIC, BIC, RMSE, MAE, MAPE, MASE
Metric %in% c("AIC","BIC","RMSE","MAE","MAPE","MASE") &
ARIMA      == min(ARIMA, ARIMAX_ALL, ARIMAX_POP, ARIMAX_REV,
   ARIMAX_GDP) ~ "ARIMA",
Metric %in% c("AIC","BIC","RMSE","MAE","MAPE","MASE") &
ARIMAX_ALL == min(ARIMA, ARIMAX_ALL, ARIMAX_POP, ARIMAX_REV,
   ARIMAX_GDP) ~ "ARIMAX_ALL",
Metric %in% c("AIC","BIC","RMSE","MAE","MAPE","MASE") &
ARIMAX_POP == min(ARIMA, ARIMAX_ALL, ARIMAX_POP, ARIMAX_REV,
   ARIMAX_GDP) ~ "ARIMAX_POP",
Metric %in% c("AIC","BIC","RMSE","MAE","MAPE","MASE") &
ARIMAX_REV == min(ARIMA, ARIMAX_ALL, ARIMAX_POP, ARIMAX_REV,
   ARIMAX_GDP) ~ "ARIMAX_REV",
Metric %in% c("AIC","BIC","RMSE","MAE","MAPE","MASE") &
ARIMAX_GDP == min(ARIMA, ARIMAX_ALL, ARIMAX_POP, ARIMAX_REV,
   ARIMAX_GDP) ~ "ARIMAX_GDP",
# Closest to zero is better for ME and MPE
Metric %in% c("ME","MPE") &
```

```
abs(ARIMA)       == min(abs(ARIMA), abs(ARIMAX_ALL),
   abs(ARIMAX_POP),
abs(ARIMAX_REV), abs(ARIMAX_GDP)) ~ "ARIMA",
Metric %in% c("ME","MPE") &
abs(ARIMAX_ALL) == min(abs(ARIMA), abs(ARIMAX_ALL),
   abs(ARIMAX_POP),
abs(ARIMAX_REV), abs(ARIMAX_GDP)) ~ "ARIMAX_ALL",
Metric %in% c("ME","MPE") & abs(ARIMAX_POP) == min(abs(ARIMA),
   abs(ARIMAX_ALL), abs(ARIMAX_POP),
abs(ARIMAX_REV), abs(ARIMAX_GDP)) ~ "ARIMAX_POP",
Metric %in% c("ME","MPE") &
abs(ARIMAX_REV) == min(abs(ARIMA), abs(ARIMAX_ALL),
   abs(ARIMAX_POP),
abs(ARIMAX_REV), abs(ARIMAX_GDP)) ~ "ARIMAX_REV",
Metric %in% c("ME","MPE") & abs(ARIMAX_GDP) == min(abs(ARIMA),
   abs(ARIMAX_ALL), abs(ARIMAX_POP),
abs(ARIMAX_REV), abs(ARIMAX_GDP)) ~ "ARIMAX_GDP",
TRUE ~ "Tie"
)) |>
ungroup()

knitr::kable(full_comparison, digits = 4,caption = "Full Model
   Comparison     All Five Specifications")
```

Table 8: Full Model Comparison — All Five Specifications

Metric	ARIMA	ARIMAX ALL	ARIMAX POP	ARIMAX REVENUE	ARIMAX GDP	BEST MODEL
AIC	-7.5107	-46.5252	13.4633	-42.9495	-21.7580	ARIMAX_ALL
BIC	-4.2889	-38.3372	18.3760	-38.0367	-15.3143	ARIMAX_ALL
ME	0.0005	0.0000	0.0000	-0.0016	0.0075	ARIMAX_ALL
RMSE	0.2044	0.1150	0.2669	0.1266	0.1593	ARIMAX_ALL
MAE	0.1684	0.0998	0.2195	0.1023	0.1270	ARIMAX_ALL
MPE	0.0043	-0.0030	-0.0165	-0.0189	0.0471	ARIMAX_ALL
MAPE	0.7819	0.4593	1.0088	0.4702	0.5882	ARIMAX_ALL
MASE	0.8428	0.4992	1.0986	0.5118	0.6357	ARIMAX_ALL

4.13.1 Model Wins Summary

```
wins <- full_comparison |>
count(Best_Model, name = "Metrics_Won") |>
arrange(desc(Metrics_Won))

knitr::kable(wins, caption = "Table 7: Number of Metrics Won
   by Each Model")
```

Table 9: Table 7: Number of Metrics Won by Each Model

Best_Model	Metrics_Won
ARIMAX_ALL	8

---

5 Chapter Five: Discussion, Conclusions and Recommendations

5.1 Discussion of Results

5.1.1 Historical Trends in Ghana's Government Expenditure

The descriptive analysis in Section 4.3 confirms that Ghana's government expenditure has grown substantially over the study period (1983–2024), exhibiting the exponential growth trajectory characteristic of a developing economy undergoing both real fiscal expansion and significant nominal exchange rate depreciation. The log transformation successfully converts this trajectory into a near-linear trend, confirming the appropriateness of the ARIMA/ARIMAX framework. Structural inflections — post-2014 cedi depreciation, COVID-19 fiscal response (2020), and the 2022–2023 debt distress episode — are visible in the transformed series and contribute to the evidence of non-stationarity at the level.

5.1.2 Stationarity and Integration Order

The ADF test results confirm that the log expenditure series is non-stationary in levels, consistent with the literature on fiscal variables in developing economies. Second-order

differencing achieves stationarity, establishing the integration order at I(2). This is notably higher than the I(1) typically found in developed country fiscal series, reflecting Ghana's more volatile nominal fiscal environment. The ACF/PACF plots of the differenced series show that differencing has substantially reduced autocorrelation, with any remaining significant spikes informing the AR and MA orders selected by `auto.arima()`.

5.1.3 Multicollinearity Among Exogenous Variables

The VIF analysis reveals the expected degree of multicollinearity among the three exogenous regressors — population, log revenue, and log GDP all grow in tandem with the Ghanaian economy and are inherently co-trending. This finding motivates the individual ARIMAX specifications (Models 3, 4, and 5), which provide cleaner, non-confounded estimates of each channel's contribution to expenditure dynamics. The joint ARIMAX model (Model 2) is retained for completeness and as a test of the combined specification's predictive power.

5.1.4 ARIMA Baseline Performance

The univariate ARIMA baseline (Model 1) captures Ghana's expenditure dynamics reasonably well, given the strong autoregressive momentum in fiscal series. However, as the comparison tables demonstrate, incorporating exogenous variables consistently improves or maintains fit quality — confirming that Ghana's expenditure path is not fully explained by its own history alone.

5.1.5 ARIMAX Model Performance and Variable Selection

The five-model comparison in Section 4.13 provides direct evidence on which exogenous variables most effectively augment the ARIMA dynamics:

- **Revenue (ARIMAX_REV)** is theoretically and empirically the most direct driver, as Ghana's expenditure closely tracks revenue availability under the cash-based budget system. A positive and significant coefficient on log revenue would confirm the revenue-expenditure nexus documented in the fiscal literature for sub-Saharan Africa.
- **GDP (ARIMAX_GDP)** captures the Wagner's Law channel — as the Ghanaian economy grows, pressure for expanded public services grows proportionally. GDP's

broader scope (capturing both private and public sector activity) makes it a useful leading indicator of fiscal expansion.

- **Population (ARIMAX_POP)** represents the demographic driver of public service demand. Ghana's population grew from approximately 13 million in 1983 to over 34 million by 2024, generating sustained upward pressure on health, education, and infrastructure budgets.

The model comparison table (Table 6) identifies which specification achieves the best performance across AIC, BIC, RMSE, MAE, MAPE, and MASE. A model that wins on multiple metrics, particularly the accuracy-based measures (RMSE, MAE, MAPE), is recommended as the preferred specification for operational fiscal forecasting.

5.1.6 Forecast Implications

The 10-year out-of-sample forecasts (Figures 7–15) project Ghana's log government expenditure forward with associated 80% and 95% prediction intervals. Key observations:

1. **Widening prediction intervals** across all models reflect compounding uncertainty in longer-horizon forecasts, which is expected and appropriate.
2. **The ARIMA baseline** produces forecasts driven solely by historical momentum and tends toward mean-reversion in the differenced space.
3. **ARIMAX forecasts** are conditioned on the mean-projection of exogenous variables. In practice, scenario-based projections (e.g., IMF WEO GDP projections, UN population forecasts) should replace the mean projection for policy-grade forecasting.
4. The forecasts suggest continued growth in government expenditure — consistent with Ghana's fiscal trajectory and demographic pressure — though the pace of growth will be sensitive to revenue performance and macroeconomic conditions.

5.2 Conclusions

This study applied ARIMAX modelling to forecast Ghana's government expenditure over the period 1984–2024, evaluating five specifications against a univariate ARIMA baseline. The following conclusions are drawn:

1. **Ghana's government expenditure is non-stationary** at the level and integrated of order 2 (I(2)), requiring second-order differencing to achieve stationarity before ARIMA-based modelling.

2. **Macroeconomic exogenous variables — population, government revenue, and GDP — meaningfully influence Ghana's government expenditure** beyond its own historical inertia, consistent with Keynesian fiscal theory and Wagner's Law.

3. **The ARIMAX model consistently outperforms the univariate ARIMA baseline** on standard forecast accuracy metrics (RMSE, MAE, MAPE, MASE), demonstrating the value of incorporating external macroeconomic information into the forecasting framework.

4. **The individual ARIMAX models** (revenue, GDP, or population as standalone exogenous variables) tend to perform comparably to or better than the joint ARIMAX model, reflecting multicollinearity among the three regressors when included simultaneously.

5. **The preferred ARIMAX specification** — identified from Table 6 — provides the most accurate and parsimonious basis for medium-term fiscal expenditure forecasting in Ghana.

6. The 10-year forecasts project continued growth in Ghana's government expenditure, consistent with population growth, economic development, and ongoing public service demand — though the pace of expansion remains sensitive to revenue mobilisation and macroeconomic stability.

5.3 Recommendations

Based on the findings, the following recommendations are made:

For the Ministry of Finance and Parliament: - Adopt the ARIMAX framework — particularly the revenue-augmented specification — as a complement to the existing in-house fiscal forecasting model. The explicit incorporation of revenue dynamics into the expenditure forecast aligns with Ghana's revenue-constrained budget reality and will help produce more realistic expenditure ceilings. - Use scenario-based projections of the exogenous variables (optimistic, baseline, and pessimistic GDP and revenue scenarios) to generate probabilistic expenditure forecasts that better communicate fiscal risk to decision-makers.

For the Bank of Ghana and National Development Planning Commission: - Maintain and regularly update the annual dataset on government expenditure, revenue, population, and GDP to enable routine re-estimation of the forecasting model as new data

become available. - Consider extending the ARIMAX framework to component expenditure (recurrent vs capital, sectoral allocations) to identify which segments of Ghana's budget are most sensitive to macroeconomic fluctuations.

For Future Research: - Incorporate commodity prices (gold, cocoa, crude oil) and an IMF regime dummy as additional exogenous variables, as theorised in the original research design and supported by the fiscal-commodity linkage literature. - Apply non-linear and machine-learning extensions (e.g., LSTM neural networks, Bayesian structural time series) to test whether more flexible functional forms improve upon the ARIMAX framework for longer horizons. - Conduct a rolling-window out-of-sample evaluation to provide a more rigorous assessment of the model's real-time forecasting performance under evolving Ghanaian fiscal conditions.

5.4 Limitations of the Study

Several limitations should be noted when interpreting the findings:

1. **Short annual sample:** With 42 observations (1983–2024), the sample limits the statistical power for detecting weak or delayed relationships, and restricts the number of parameters that can be reliably estimated.
2. **Mean exogenous projections:** The use of in-sample mean values to project future exogenous variables is a simplification. More realistic scenario-based projections would improve the policy relevance of the forecasts.
3. **Parameter instability:** Ghana's economy has undergone several structural changes (oil discovery in 2010, multiple IMF programmes, debt restructuring in 2022–2023) that may have altered the underlying fiscal reaction function. Structural break tests and time-varying parameter models were not applied in this study.
4. **Omitted variables:** The model does not capture all drivers of Ghana's expenditure — commodity price cycles, IMF conditionality effects, election cycles, and aid flows are potentially important omissions that future work should address.

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7 Appendix: Session Information

```r
sessionInfo()
```

```
R version 4.5.2 (2025-10-31 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 11 x64 (build 26200)

Matrix products: default
LAPACK version 3.12.1

locale:
[1] LC_COLLATE=English_United Kingdom.utf8  LC_CTYPE=English_United Kingdom.utf8
LC_MONETARY=English_United Kingdom.utf8
[4] LC_NUMERIC=C                            LC_TIME=English_United Kingdom.utf8

time zone: Africa/Accra
tzcode source: internal

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base

other attached packages:
[1] lubridate_1.9.4 forcats_1.0.1   stringr_1.5.2   dplyr_1.1.4     purrr_1.0.4
readr_2.1.5     tidyr_1.3.1     tibble_3.2.1    ggplot2_4.0.0
[10] tidyverse_2.0.0 readxl_1.4.5    car_3.1-3       carData_3.0-6   tseries_0.10-58 forecast_9.0.1

loaded via a namespace (and not attached):
[1] utf8_1.2.6         generics_0.1.4     stringi_1.8.7      lattice_0.22-7
hms_1.1.4          digest_0.6.37      magrittr_2.0.3     timechange_0.3.0
[9] evaluate_1.0.5     grid_4.5.2         RColorBrewer_1.1-3 fastmap_1.2.0
cellranger_1.1.0   Formula_1.2-5      scales_1.4.0       abind_1.4-8
[17] cli_3.6.4          rlang_1.1.6        withr_3.0.2        yaml_2.3.10
tools_4.5.2        parallel_4.5.2     tzdb_0.5.0         colorspace_2.1-2
[25] curl_6.2.2         vctrs_0.6.5        R6_2.6.1           zoo_1.8-14
lifecycle_1.0.4    pkgconfig_2.0.3    urca_1.3-4         pillar_1.11.1
```

```
[33] gtable_0.3.6       rsconnect_1.5.1    glue_1.8.0         quantmod_0.4.28
Rcpp_1.0.14        xfun_0.53          tidyselect_1.2.1   rstudioapi_0.17.1
[41] knitr_1.50         farver_2.1.2       htmltools_0.5.8.1  nlme_3.1-168
labeling_0.4.3     rmarkdown_2.30     xts_0.14.1         timeDate_4051.111
[49] fracdiff_1.5-3     compiler_4.5.2     S7_0.2.0           quadprog_1.5-8     TTR_0.24.4
```